Nonlinear Analysis: Hybrid Systems 52 (2024) 101455

Contents lists available at ScienceDirect

Nonlinear Analysis: Hybrid Systems

journal homepage: www.elsevier.com/locate/nahs

Duality of controllability and observability in proportional equal conflict timed continuous Petri Nets

J.L. García-Malacara [a], César Arzola [b], Antonio Ramírez-Treviño [a,*], C. Renato Vázquez [c]

[a] CINVESTAV, Av. del Bosque 1145, Zapopan, 45019, Jalisco, Mexico
[b] Departamento de Informática e Ingeniería de Sistemas, María de Luna 1, Zaragoza, 50018, Spain
[c] Tecnologico de Monterrey, Av. Ramón Corona 2514, Zapopan, 45201, Jalisco, Mexico

ARTICLE INFO

Keywords:
Timed continuous Petri Nets
Controllability
Observability
Duality
Application of nonlinear analysis and design

ABSTRACT

Controllability and observability properties have been widely studied in Timed Continuous Petri Nets ($TCPN$s), a class of *piecewise affine systems*, in order to analyze and control crowded discrete event systems. This work studies the concept of duality applied to $TCPN$s as a vehicle to establish links between controllability and observability, i.e., a synergy to improve the understanding of these properties and to enlarge the class of nets that can be analyzed. To achieve this, we study the concepts of rank-controllability and rank-observability. They capture structural conditions for controllability and observability. Afterwards, the computation of dual nets for Fork-Attribution (FA), Choice-Free (CF), Join-Free (JF), and Proportional Equal Conflict (PEQ) $TCPN$s subclasses are presented. By using the dual definition, several relations between the primal's controllability and its dual's observability are stated. Particularly, in FA rank-controllability and rank-observability are dual properties. In consistent and strongly connected CF, JF, and PEQ nets, the rank-observability of the dual is sufficient for the rank-controllability of the primal. The opposite implication holds for CF and PEQ if the self-loop places, added by the dual construction methodology, are measurable.

1. Introduction

Finite Automata and Petri Nets (PNs) are two formalisms widely used to represent and analyze Discrete Event Systems (DES), such as power systems [1], flexible manufacturing systems [2], traffic systems [3,4], among many others. In this paper, we are interested in the study of *controllability* and *observability*, which are two dynamical properties of these systems that have been largely studied since they deal with the possibility of confining the system into a required behavior [5,6], and estimating the internal system state [7,8]. Nevertheless, these formalisms are limited when dealing with the analysis of modern DESs, since these systems are large and complex, processing huge amounts of products and/or information, leading to the well-known *state explosion problem*. In order to overcome this limitation, *fluidization* is proposed in [9] as a relaxation technique by which the firing of the transitions is allowed to be a non-negative real amount and consequently, the marking becomes a non-negative real amount. Moreover, by associating time rates to transitions, the notion of time is introduced, resulting in the (transition) Timed Continuous Petri Nets ($TCPN$s) model. Different time interpretations can be used for these nets; through this work, the *infinite server semantics* (ISS) is used [10]. They represent an interesting and powerful tool since they can be used to approximate timed DES models [11,12] and function as

* Corresponding author.
E-mail addresses: j.l.g.malacara@tec.mx (J.L. García-Malacara), carzola@unizar.es (C. Arzola), antonio.ramirezt@cinvestav.mx (A. Ramírez-Treviño), cr.vazquez@tec.mx (C.R. Vázquez).

https://doi.org/10.1016/j.nahs.2023.101455
Received 23 January 2023; Received in revised form 20 July 2023; Accepted 1 December 2023
Available online 15 December 2023

standalone models [13]. Moreover, under ISS, $TCPN$s become a particular class of piecewise affine systems (PWA) [14], thus the results herein introduced may easily be translated to this class of systems.

For the particular case of $TCPN$s under ISS, the analysis of controllability and observability has been a significant area of research and study [15–21]; when a $TCPN$ is controllable, there exists a controller leading the system from the initial marking to any other one (for instance the marking with maximum throughput), by dynamically slowing down the maximal execution speed at some particular transitions (representing events), named controllable; when a $TCPN$ is observable, its complete state (i.e., the global state) can be estimated based on the knowledge of the marking at some particular places (local states).

In [21] the study of controllability was divided into two cases, when all the transitions are controllable and when there exist uncontrollable transitions. In the first case, the *consistency* of the net is sufficient and necessary to guarantee the $TCPN$ controllability. In the second case, the controllability characterization, based on the controllability matrix, was limited to sets of equilibrium markings, defined a priori. In [20], structural and generic conditions for controllability of some $TCPN$ subclasses were presented. In these works, the controllability analysis has been performed by analyzing the *configurations* of the net. By following a different approach, in [17], the concept of *rank-controllability*, which only relies on the structural conditions of the net, is used to determine when a $TCPN$ is controllable. It is useful to avoid the analysis by configurations, however, the conditions for controllability established by this method are restrictive in the sense that they require that almost all transitions within the $TCPN$ must be controllable.

Regarding observability, in [18] necessary conditions for observability in the JF were provided. In [19], the concept of *distinguishable modes* and generic observability in $TCPN$s were introduced, and sufficient structural conditions for generic observability were provided. Alternatively, in [16] the observability of $TCPN$s was studied by using a geometric approach, structurally characterizing certain unobservable subspaces. In [15] a sensor placement problem to guarantee the distinguishability of $TCPN$s was addressed.

It is worth noticing that, on one hand, observability is characterized as a global property, i.e. whether the net is observable for all the configurations or not; while controllability with uncontrollable transitions is characterized, in the vast majority of works, as a local property, whose analysis is performed for each configuration. In other words, the observability characterizations do not demand the knowledge of configurations, as in the case of controllability characterizations; the enumeration of all the configurations will lead to an NP-Complete problem. On the other hand, the controllability characterization deals with the construction of controllability matrices fulfilling certain properties, i.e. an algebraic approach, considering particularities as the input boundedness and the locality of the property, being possible to use approaches and techniques from the continuous systems control theory.

This work proposes to build a bridge between controllability and observability. It allows taking advantage of both characterizations, in order to derive more efficient and precise results and to obtain a wider understanding of the connection between the net structure and the possible transient behaviors in $TCPN$s. In order to do that, the concept of controllability - observability duality of $TCPN$s introduced in [22] is used to relate those properties.

First, a procedure for the computation of the dual net of a Fork-Attribution (FA) is introduced. In it, the token-flow is reversed, based on the definition for discrete nets introduced in [23]. Later, procedures for the computation of the dual net in Choice-Free (CF), Join-Free (JF), and Proportional Equal Conflict (PEQ) subclasses are derived. For CF nets the input arcs to a join transition that are not constraining its flow (i.e. at a certain configuration) are reversed. For JF nets, non controllable transitions in conflict are merged with an algorithm herein proposed. Both operations are required for PEQ nets. After removing join transitions and choices, nets resemble FA $TCPN$s, hence the procedure for computing the dual of a FA $TCPN$ can be used. Therefore, a dual net is computed for each configuration of the primal. In order to deal with systems with a large number of configurations, herein the concept of dual-composition is introduced, which includes all the duals of a net in a single structure.

Based on duality, three novel results relating controllability and observability are presented. In order to do this, we use the concepts of rank-controllability and introduce the concept of *rank-observability*; they establish structural conditions for controllability and observability, respectively. Then, the first result states that the rank-controllability in a consistent and strongly connected FA $TCPN$ is equivalent to the rank-observability of its dual net, and vice versa. The second result states that, for a given configuration, if the dual of a consistent and strongly connected CF $TCPN$ is rank-observable then the primal net is rank-controllable, the converse holds if additionally, the self-loop places of the dual are measurable. Lastly, the third result builds upon the previous findings and presents the relation between rank-observability and rank-controllability for PEQ $TCPN$ systems.

This work is organized as follows. Section 2 presents the PN and $TCPN$ models. In Section 3 some concepts and results about controllability and observability in $TCPN$s are recalled, and the definitions of rank-controllability and rank-observability are presented. In Section 4, the adaptation of the classic duality concept to the class of FA TCPN systems is introduced. Moreover, the relationship between rank-observability and rank-controllability is presented. These results are extended to the CF class in Section 5, to the JF class in Section 6 and to the PEQ class in Section 7. An application example is shown in Section 8. Finally, conclusions and future works are presented in Section 9.

2. Basic concepts

2.1. Petri nets

This subsection introduces the Petri net model. An interested reader should refer to [9,10] for a deeper insight into this field. Through this work, the following notation will be used. Lowercase bold letters represent vectors, for instance, $\mathbf{a}$ is a vector and a_i is its ith entry. Similarly, upper case bold letters represent matrices, for instance, $\boldsymbol{A}$ is a matrix, and A_{ij} is its element in the ith row and jth column; $\boldsymbol{A}(i,\bullet)$ and $\boldsymbol{A}(\bullet,j)$ denote the ith row and jth column, respectively. The support of a vector $\mathbf{a}$ is denoted as $\|\mathbf{a}\|$ and it is defined as the indexes of the non-null entries of $\mathbf{a}$. The notation $\mathbf{a} > \mathbf{b}$ means that $a_i > b_i$ for every i; $\mathbf{a} \gneq \mathbf{b}$ means that $a_i \geq b_i$ for every i and for at least for one i, $a_i > b_i$.

Definition 1. A Petri net structure ($\mathcal{N}$) is a bipartite digraph formed by the four-tuple $\mathcal{N} = \langle P, T, \boldsymbol{Pre}, \boldsymbol{Post}\rangle$ where $P = \{p_1, p_2, \ldots, p_{|P|}\}$ is a finite set of nodes called places; $T = \{t_1, t_2, \ldots, t_{|T|}\}$ is a finite set of nodes called transitions; $P \cap T = \emptyset$. $\boldsymbol{Pre}$ and $\boldsymbol{Post}$ are $|P| \times |T|$ matrices with nonnegative rational entries. The i,jth entry of matrix $\boldsymbol{Pre}$, Pre_{ij}, ($\boldsymbol{Post}$, $Post_{ij}$,) represents the weight of arc going from place p_i to transition t_j (from transition t_j to place p_i).

When all entries Pre_{ij}, $Post_{ij}$ are zero or one, the net is said to be *ordinary*. The *token-flow* matrix of $\mathcal{N}$ is defined as $\boldsymbol{C} = \boldsymbol{Post} - \boldsymbol{Pre}$. Graphically, places, transitions and arcs are represented by circles, rectangles, and arrows pointing from the source node to the ending one, respectively. Let $P' \subset P$ and $T' \subset T$, then $\boldsymbol{Pre}(P', T')$ denotes the restriction of $\boldsymbol{Pre}$ to rows P' and columns T', similarly for $\boldsymbol{Post}(P', T')$.

Let $x \in P \cup T$ be a node of $\mathcal{N}$. The input set of x, denoted by ${}^\bullet x$, is defined as ${}^\bullet x = \{x_i \in P \cup T|$ there exists an arc from x_i to $x\}$. Similarly, the output set of a node x, denoted by $x^\bullet$, is defined as $x^\bullet = \{x_i \in P \cup T|$ there exists an arc from x to $x_i\}$. A *T-Semiflow* (*P-Semiflow*) is a rational solution $\mathbf{x} \gneq \mathbf{0}$ ($\mathbf{y} \gneq \mathbf{0}$) for $\boldsymbol{C} \cdot \mathbf{x} = \mathbf{0}$ ($\mathbf{y}^T \cdot \boldsymbol{C} = \mathbf{0}$). If there exists a T-Semiflow $\mathbf{x} > \mathbf{0}$ then $\mathcal{N}$ is said to be *consistent* (C_t). If there exists a P-Semiflow $\mathbf{y} > \mathbf{0}$ then $\mathcal{N}$ is said to be *conservative* (C_v). If $\mathbf{x} \neq \mathbf{0}$ (resp. $\mathbf{y} \neq \mathbf{0}$) is a rational solution of $\boldsymbol{C} \cdot \mathbf{x} = \mathbf{0}$ (resp. $\mathbf{y}^T \cdot \boldsymbol{C} = \mathbf{0}$) then it is named *T-flow* (resp. *P-flow*). A basis for the left null space of $\boldsymbol{C}$ is denoted by $\boldsymbol{B}_y^T$, and a basis for the right null space of $\boldsymbol{C}$ is denoted by $\boldsymbol{B}_x$.

A subnet $\mathcal{N}'$ of $\mathcal{N}$ is defined as $\mathcal{N}' = \left\langle P', T', \boldsymbol{Pre}', \boldsymbol{Post}'\right\rangle$ where $P' \subset P$, $T' \subset T$, $\boldsymbol{Pre}' = \boldsymbol{Pre}(P', T')$ and $\boldsymbol{Post}' = \boldsymbol{Post}(P', T')$. A *P-subnet* induced by a P-semiflow $\mathbf{y}$, denoted $\mathcal{P}_y$, is the subnet $\mathcal{P}_y = \langle P', T', \boldsymbol{Pre}', \boldsymbol{Post}'\rangle$ where $P' = \{p_i | i \in \|\mathbf{y}\|\}$, $T' = {}^\bullet P' \cup P'^\bullet \subset T$, $\boldsymbol{Pre}' = \boldsymbol{Pre}(P', T')$ and $\boldsymbol{Post}' = \boldsymbol{Post}(P', T')$.

A *PN* is *strongly connected* if for every pair of nodes v and w in the net, there are paths leading from v to w, and from w to v. Two transitions t and t' are in *conflict* if ${}^\bullet t \cap {}^\bullet t' \neq \emptyset$. Two transitions t and t' are in *proportional equal conflict* (PEQ) if $\exists \gamma > 0$ s.t. $\boldsymbol{Pre}(\bullet, t) = \gamma \boldsymbol{Pre}(\bullet, t')$.

The concept of dual nets in PNs is used through this work.

Definition 2 (*[23]*)**.** Let $\mathcal{N}^p$ be a Petri net structure, named the *primal* net. Its *dual* is another Petri net structure, denoted as $\mathcal{N}^d$, where the sets of places and transitions are interchanged and the direction of arcs are reversed.

Notice that primal and dual incidence matrices, denoted as $\mathbf{C}^p$ and $\mathbf{C}^d$ respectively, are related by $\mathbf{C}^d = (\mathbf{C}^p)^T$. The analysis of properties exhibited by the dual net can be exploited to derive properties in the primal net. For instance, [23] presented results where well-formedness, repetitiveness, and boundedness were derived using properties of the dual.

2.2. Continuous Petri nets

A continuous Petri net is a Petri net structure $\mathcal{N}$ together with a marking, where transitions can be activated in any real amount between zero and its *enabling degree*. As a consequence, the number of tokens residing in a place is represented by a non-negative real number. This model is formally defined below.

Definition 3. A *continuous* Petri net (CPN) is the tuple $\left\langle \mathcal{N}, \boldsymbol{m}_0\right\rangle$ where $\mathcal{N}$ is the Petri net structure and $\boldsymbol{m}_0$ is the initial marking, where the marking $\boldsymbol{m} : P \to \mathbb{R}_{\geq 0}^{|P|}$ is a vector representing the non-negative real number of tokens residing inside each place.

In a CPN a transition $t_i \in T$ is *enabled* iff for every $p_j \in {}^\bullet t_i$, $\boldsymbol{m}(p_j) > 0$. The *continuous enabling degree* of a transition t_i is given by

$$enab(t_i, \boldsymbol{m}) = \min_{p_j \in {}^\bullet t_i} \left\{ \frac{\boldsymbol{m}(p_j)}{\boldsymbol{Pre}(p_j, t_i)} \right\} \tag{1}$$

The enabling degrees are well defined if every transition has at least one input place, which is assumed through this work. An enabled transition t_i at a marking $\boldsymbol{m}$ can be fired in any real amount $0 \leq \sigma_i \leq enab(t_i, \boldsymbol{m})$, leading to a new marking $\boldsymbol{m}'$ that is computed using the fundamental CPN equation:

$$\boldsymbol{m}' = \boldsymbol{m} + \boldsymbol{C}\vec{\sigma} \tag{2}$$

where $\vec{\sigma} = \begin{bmatrix}\sigma_1 & \ldots & \sigma_{|T|}\end{bmatrix}^T$.

The set of all reachable markings in the CPN is denoted as $R(\langle \mathcal{N}, \boldsymbol{m}_0\rangle)$.

2.3. Timed continuous Petri nets

Definition 4. A *timed continuous Petri net (TCPN)* (or timed fluid net) is a time-driven continuous-state system described by the tuple $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0\rangle$, where $\langle \mathcal{N}, \boldsymbol{m}_0\rangle$ is a *continuous* PN system and the vector $\lambda \in \mathbb{R}_{>0}^{|T|}$ represents the transition firing rates.

In a $TCPN$ transitions fire according to certain speed, which generally is a function of their firing rates and the instantaneous marking. Analogously to stochastic timed PNs, under *infinite server semantics*, an enabled transition t_i fires with a speed defined as the product of its associated rate, λ_i, and its enabling degree, $enab(t_i, \boldsymbol{m})$. The *flow vector* $\mathbf{f(m)}$ is defined such that $f_i(\mathbf{m})$ denotes the flow through t_i, thus,

$$f_i(\boldsymbol{m}) = \lambda_i \cdot enab(t_i, \boldsymbol{m}) = \lambda_i \cdot \min_{p_j \in {}^\bullet t_i} \left\{ \frac{\boldsymbol{m}(p_j)}{\boldsymbol{Pre}(p_j, t_i)} \right\}.$$

Given a $TCPN$ $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$, we will refer to the pair $\langle \mathcal{N}, \lambda \rangle$ as a $TCPN$ timed structure. The enabling degree vector $\mathbf{enab}(\boldsymbol{m}) \in \mathbb{R}_{\geq 0}^{|T|}$ is defined such that $enab_i(\boldsymbol{m}) = enab(t_i, \boldsymbol{m})$.

Definition 5. A *configuration* is a set of arcs $C_j = \{(p_i, t_k) | p_i \in {}^\bullet t_k\}$ where every $t_k \in T$ appears in only one pair. Given a configuration, C_j, if $(p_i, t_k) \in C_j$, it is said that p_i *constrains* the flow of t_k at C_j. An upper bound for the number of configurations is $\prod_{t \in T} |{}^\bullet t| = |{}^\bullet t_1| \times \cdots \times |{}^\bullet t_{|T|}|$. For each possible configuration C_j, the configuration matrix $\boldsymbol{\Pi}_j$ of dimension $|T| \times |P|$ is defined as:

$$\forall i \in \{1, \ldots, |T|\}, \forall k \in \{1, \ldots, |P|\}$$
$$\boldsymbol{\Pi}_{j\,ik} = \begin{cases} \frac{1}{\boldsymbol{Pre}(p_k, t_i)}, & \text{if } (p_k, t_i) \in C_j \\ 0, & \text{otherwise} \end{cases}$$

Then, the flow through the transitions in a given configuration C_j can be written as the vector $\mathbf{f}(\boldsymbol{m}) = \boldsymbol{\Lambda \Pi}_j \boldsymbol{m}$, where $\boldsymbol{\Lambda}$ is the diagonal matrix whose elements Λ_{ii} are the firing transition rates λ_i.

In the systems modeled by $TCPN$s, control actions can be seen as decrements of activity speeds from the maximum allowed (uncontrolled) activity speed (in machines, for instance, their throughput can only be slowed down, from their maximal throughput value). Thus, control actions in $TCPN$s are applied to transitions, decreasing their flow. Transitions where a control action can be applied are said to be *controllable*. The set of all controllable transitions is denoted by T_c, and the set of uncontrollable transitions is denoted by $T_{nc} = T \setminus T_c$.

The control vector $\mathbf{u} \in \mathbb{R}^{|T|}$ is defined such that u_i represents the control action over $t_i \in T$. The effective flow through a controlled transition is given by:

$$\begin{aligned} & w_i(\boldsymbol{m}) = f_i(\boldsymbol{m}) - u_i, \\ & \text{with } 0 \leq u_i \leq \lambda_i \cdot enab_i(\boldsymbol{m}), \end{aligned} \tag{3}$$

thus $w_i(\boldsymbol{m}) \geq 0$. Moreover, if $t_i \in T_{nc}$ then, by definition, u_i must be equal to zero. A $TCPN$ with an input control is named *forced*. When the value of u_i (associated to a $t_i \in T_c$) meets the constraint equation $0 \leq u_i \leq \lambda_i \cdot enab_i(\boldsymbol{m})$, then it is said that the flow w_i is controlled by a *suitably bounded control input* u_i.

Moreover, this work considers that only the value of the marking at some places $P_m \subseteq P$, named *measurable*, is available for an external observer.

Thus, the behavior of a $TCPN$ system, in any configuration C_i is described by the state equation:

$$\begin{aligned} \dot{\boldsymbol{m}} &= \boldsymbol{C \Lambda \Pi}_i \boldsymbol{m} - \boldsymbol{C}\mathbf{u} \\ &= \boldsymbol{C \Lambda \Pi}_i \boldsymbol{m} - \boldsymbol{C}_c \mathbf{u}_c, \qquad \text{with } \mathbf{0} \leq \mathbf{u} \leq \boldsymbol{\Lambda \Pi}_i \boldsymbol{m} \\ \mathbf{z} &= \boldsymbol{O m} \end{aligned} \tag{4}$$

where $\boldsymbol{C \Lambda \Pi}_i$ is the *dynamic matrix* for C_i; the *input matrix* $\boldsymbol{C}_c = \boldsymbol{C}(P, T_c)$ is the restriction of the incidence matrix $\boldsymbol{C}$ to the columns of controllable transitions; $\mathbf{u}_c = \mathbf{u}(T_c)$ is the restriction of $\mathbf{u}$ to the controllable transitions T_c; $\boldsymbol{O}$ is the *output matrix*, which is a selector of the measurable places, i.e., a $|P_m| \times |P|$ matrix s.t. if $p_k \in P_m$, the $|P|$-sized elementary vector $\mathbf{e}_k$ is a row of $\boldsymbol{O}$ ($\mathbf{e}_k$ is a vector of zeros whose kth entry is 1); $\mathbf{z}$ is the $TCPN$ *output vector*. Notice that $\boldsymbol{C}_c = \boldsymbol{C} \cdot \boldsymbol{S}$ where $\boldsymbol{S}$ is a selector of controllable transitions, i.e. a $|T| \times |T_c|$ matrix, $\boldsymbol{S}$, s.t., if $t_k \in T_c$, the $|T|$-sized elementary vector $\mathbf{e}_k$ is a column of $\boldsymbol{S}$[1] The set of all reachable markings (i.e. those that are solutions of (4)) is denoted as $R(\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle)$.

Definition 6. The set $Class(\boldsymbol{m}_0) = \left\{ \boldsymbol{m} \in \mathbb{R}_{\geq 0}^{|P|} | \mathbf{B}_y^T \boldsymbol{m} = \mathbf{B}_y^T \boldsymbol{m}_0 \right\}$ contains all the markings agreeing with $\boldsymbol{m}_0$ in the P-flows. Notice that $Class(\boldsymbol{m}_0) \supseteq R(\langle \mathcal{N}, \boldsymbol{m}_0 \rangle) \supseteq R(\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle)$.

2.4. Petri nets subclasses

Through this work, the following Petri net structural subclasses are considered.

Definition 7. Subclasses of nets defined according to their structure:

1. $\mathcal{N}$ is a State Machine (SM) if it is ordinary and $\forall t \in T, |{}^\bullet t_i| = |t_i^\bullet| = 1$.
2. $\mathcal{N}$ is a Join-Free (JF) if $\forall t \in T, |{}^\bullet t_i| \leq 1$.
3. $\mathcal{N}$ is a Choice-free (CF) if $\forall p \in P, |p^\bullet| \leq 1$.

[1] To guarantee the uniqueness of matrix $\boldsymbol{O}$ (resp., $\boldsymbol{S}$) the indices of its rows (resp., columns), $\mathbf{e}_k$, are in ascending order.

4. $\mathcal{N}$ is a Fork-Attribution (FA) if it is CF and JF.
5. $\mathcal{N}$ is a Marked Graph (MG) if it is ordinary and $\forall p \in P, |^{\bullet}p| = |p^{\bullet}| = 1$.
6. $\mathcal{N}$ is a Proportional Equal Conflict (PEQ) if all conflicts are proportionally equal, i.e., for each pair of transitions t, t' in conflict it holds $\exists \gamma > 0$ s.t. $\boldsymbol{Pre}(\bullet, t) = \gamma \boldsymbol{Pre}(\bullet, t')$.

One of the advantages of the study of subclasses of nets is that, usually, the results obtained to analyze behavioral properties in general systems can be improved for the subclasses under consideration. For instance, while for general nets only structural necessary or sufficient conditions for liveness exist, for PEQ systems, structural necessary and sufficient conditions can be stated [10](which can be verified in polynomial time).

JF systems are a generalization of the well-known subclass of State Machines. CF nets represent systems without structural conflicts [24]; in particular, they comprise the subclass of Marked Graphs. JF and CF nets belong to the class of continuous PEQ nets [10]. These nets are able to capture the *cooperative* and *decision* processes occurring in a manufacturing system. Cooperative processes represent resource allocation to jobs and assembly operations; decision processes represent alternative processing routes followed by parts. It leads that PEQs are able to represent a broad subclass of *flexible manufacturing systems* (FMS) that includes the Flow Shop and Job Shops, among other FMS subclasses, where the decision on which route a part may take, is not subordinated to the availability of idle resources. In Section 8, we present an illustrative example applying the proposed method to a FMS, showcasing the practicality of the proposed approach to real-life systems.

The following two propositions for CF nets will be used in the subsequent developments.

Proposition 1 (*Theorem 8 [24]*)**.** *Let $\mathcal{N}$ be a strongly connected CF net:*

1. *$\mathcal{N}$ is consistent iff it has a T-semiflow.*
2. *$\mathcal{N}$ has at most one minimal T-semiflow.*

Proposition 2 (*Theorem 12 [24]*)**.** *Let $\mathcal{N}$ be a consistent CF net:*

1. *$\mathcal{N}$ conservative iff it is strongly connected.*

Proposition 3 (*Lemma 11 [24]*)**.** *Let $\mathcal{N}$ be a strongly connected FA net:*

1. *$\mathcal{N}$ is bounded iff it has a P-semiflow.*
2. *$\mathcal{N}$ has at most one minimal P-semiflow.*
3. *$\mathcal{N}$ conservative iff it is consistent.*

Notice that, in JF nets, $|P| \leq |T|$, while in CF nets, $|T| \leq |P|$. Then, in FA nets, $|P| = |T|$. Moreover, the dynamical behavior of a JF net is described by a positive linear system. The FA subclass is closed under the dual operation of Definition 2, in the sense that the dual for a FA is another FA. In other subclasses there exist joins and/or choices, whose dual structures are choices and joins, respectively. Nevertheless, joins introduce non-linearities in the $TCPN$ evolution, a phenomenon that does not occur in choices. Thus, the dual concept in general $TCPN$s should involve more information than that provided by the structure.

3. Controllability and observability of $TCPN$s

In this section, the concepts of observability and controllability in $TCPN$s are recalled. Moreover, the concept of *rank-controllability*, introduced in [17], and the novel concept of *rank-observability*, are presented; they represent conditions for controllability and observability in a given configuration, respectively. In the rest of the paper, the duality concept will be used as a bridge to translate the rank-observability characterization of a system in terms of the rank-controllability and vice versa.

3.1. Observability concepts

Definition 8 (*[19]*)**.** Let $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ be a $TCPN$ system and $P_o \subset P$ be the set of measurable places. $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ is said to be *observable in infinitesimal time* if it is possible to compute its initial marking from the knowledge of the marking of the output places at any time in interval $[0, \epsilon)$, $\forall \epsilon > 0$.

Since the $TCPN$ in a given configuration C_i is a linear system, the observability can be analyzed by using the classical *observability matrix*, which is defined as follows

$$\mathcal{O}_i = \begin{bmatrix} \boldsymbol{O} \\ \boldsymbol{O} \cdot \boldsymbol{C\Lambda\Pi}_i \\ \vdots \\ \boldsymbol{O} \cdot (\boldsymbol{C\Lambda\Pi}_i)^{|P|-1} \end{bmatrix} \tag{5}$$

The observability matrix characterizes the observability of the corresponding configuration. The following concept is introduced regarding the observability matrix.

Definition 9. A $TCPN$ timed structure $\langle \mathcal{N}, \lambda \rangle$ is said to be *rank-observable at configuration* C_i if $rank(\mathcal{O}_i) = |P|$.

The following proposition can be directly derived from Theorem 15 of [19]. It relates the rank-observability property with the observability in a $TCPN$.

Proposition 4. *Let $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ be a $TCPN$. If the system is evolving at a known configuration C_i, $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ is observable in infinitesimal time iff $\langle \mathcal{N}, \lambda \rangle$ is rank-observable in C_i.*

3.2. Controllability concepts

Since the $TCPN$ in a given configuration C_i is a linear system, the controllability matrix is defined as follows (see [25]):

$$\mathscr{C}_i = \begin{bmatrix} \boldsymbol{CS} & \boldsymbol{C\Lambda\Pi}_i\boldsymbol{CS} & \dots & \boldsymbol{C\Lambda\Pi}_i^{|P|-1}\boldsymbol{CS} \end{bmatrix} \tag{6}$$

Then $rank(\mathscr{C}_i)$ is at most $rank(\boldsymbol{C})$. Hence, conservative $TCPN$ systems (in which $\exists \mathbf{y} > \mathbf{0}$ s.t. $\mathbf{y}^T\mathbf{C} = \mathbf{0}$, thus $rank(\mathbf{C}) < |P|$) are uncontrollable according to the classical controllability definition for linear systems. In order to overcome this problem, the controllability in $TCPN$s has been studied in the past over sets of equilibrium markings (i.e. the markings where the system can be maintained by means of a suitably bounded control input $\mathbf{u}$). The following concepts are taken from [21]. First, let us recall the concept of equilibrium points.

Definition 10. Let $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ be a $TCPN$ system. Assume that it is evolving in configuration C_i. The point $(\boldsymbol{m}, \mathbf{u})$ is an *equilibrium point of configuration* C_i if $\dot{\boldsymbol{m}} = \boldsymbol{C\Lambda\Pi}_i\boldsymbol{m} - \boldsymbol{C}_c\mathbf{u} = \mathbf{0}$, with $0 \le u_j \le \lambda_j enab_j(\mathbf{m})$, $\forall t_j \in T_c$, and $u_j = 0$, $\forall t_j \in T_{nc}$. The set of *interior equilibrium points in configuration* C_i is defined as

$$\mathbf{E}_i^* = \{(\boldsymbol{m}, \mathbf{u}) | (\boldsymbol{m}, \mathbf{u}) \text{ is an equilibrium point of } C_i \text{ and } 0 < \mathrm{u}_j < f_j(\boldsymbol{m}), \forall t_j \in T_c\}$$

Notice that $\mathbf{E}_i^*$ does not include the equilibrium points that belong the borders of $Class(\boldsymbol{m}_0)$. It is because in the border some control actions could be blocked due to that some places could reach zero marking. The following definition introduces the controllability concept in $TCPN$s.

Definition 11. Let $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ be a $TCPN$ system evolving in configuration C_i. The $TCPN$ $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ is *controllable over* $\mathbf{E}_i^* \subset Class(\boldsymbol{m}_0)$ if for any two markings $\boldsymbol{m}_1$, $\boldsymbol{m}_2 \in \mathbf{E}_i^*$ there exists a suitably bounded input $\mathbf{u}(\tau)$ transferring the system described by (4) from $\boldsymbol{m}_1$ to $\boldsymbol{m}_2$.

The controllability has been analyzed for the subsets $\mathbf{E}_i^*$ defined for each configuration of the $TCPN$, determining if the system can be moved and maintained on any marking in the set of equilibrium points. Some sufficient conditions for controllability were provided in [21], however, some practical difficulties were found. In particular, the sets of equilibrium markings have to be characterized for each configuration; moreover, it is assumed that the union of the sets of equilibrium markings is connected.

In order to establish a link between controllability and observability through the net duality, the aspects of controllability related to structural and marking-dependent (behavioral) characteristics must be separated. As mentioned above, the timed structure of the $TCPN$, in each configuration, determines the controllability matrix C_i whose rank is at most $rank(\boldsymbol{C})$. The following structural concept captures the previous ideas. From now, let us denote as $Im(\bullet)$ the image of the argument matrix.

Definition 12 (*[17]*)**.** A $TCPN$ timed structure $\langle \mathcal{N}, \lambda \rangle$ is said to be rank-controllable in a configuration C_i if $rank(\mathscr{C}_i) = |P| - rank(\boldsymbol{B}_y^T) = rank(\boldsymbol{C})$. In such case, $Im(\mathscr{C}_i) = Im(\boldsymbol{C})$.

The following proposition establishes a relation between rank-controllability and controllability over the corresponding E_i^*.

Proposition 5 (*[17]*)**.** *Let $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ be a $TCPN$, where $\boldsymbol{m}_0 \in E_i^*$. If $\langle \mathcal{N}, \lambda \rangle$ is rank-controllable in C_i then it is controllable over E_i^*.*

By analyzing the rank-controllability of a $TCPN$ system, structural controllability conditions can be derived without specifying equilibrium markings or initial markings, thus leading to more efficient techniques. Let us illustrate the concepts introduced in this subsection.

Example 1. Consider the $TCPN$ depicted in Fig. 1(a). There are two possible configurations $C_1 = \{(p_1, t_2), (p_2, t_1), (p_4, t_3)\}$ and $C_2 = \{(p_3, t_2), (p_2, t_1), (p_4, t_3)\}$. Let the set of controllable transitions be $T_c = \{t_1\}$, and $\lambda = [1, 1, 1]^T$. The incidence and configuration matrices are:

$$\boldsymbol{C} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}, \quad \boldsymbol{\Pi}_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{\Pi}_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{7}$$

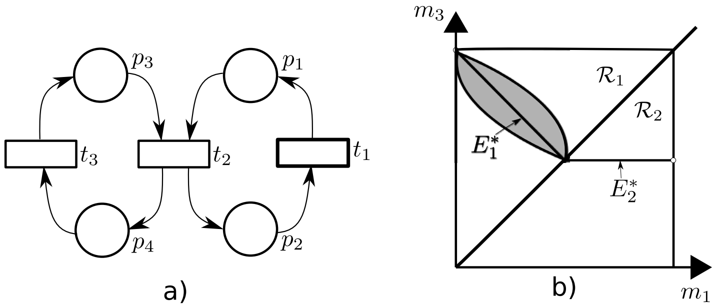

Fig. 1. (a) $TCPN$ of Example 1 and, (b) the equilibrium markings E_1^* and E_2^* in each configuration, which along with the gray area form the reachability set. Since $m_1 + m_2 = k_1$; $m_3 + m_4 = k_2$, hence the reachability space can be projected into coordinates m_3 vs m_1, as in (b); the reachability space is given by $\mathcal{RS} = \mathcal{R}_1 \cup \mathcal{R}_2$, where $\mathcal{R}_1$ is the reachability set when $m_3 \geq m_1$ and $\mathcal{R}_2$ is the reachability set when $m_1 \geq m_3$. If $m_1 = m_3$ then any configuration can be considered as valid.

For the two configurations, the dynamic matrices $\boldsymbol{C\Lambda\Pi}_i$ are:

$$\boldsymbol{C\Lambda\Pi}_1 = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \end{bmatrix}, \quad \boldsymbol{C\Lambda\Pi}_2 = \begin{bmatrix} 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}. \tag{8}$$

It is easy to check that $rank(\boldsymbol{C}) = 2$, and the ranks of the controllability matrices are $rank(\mathscr{C}_1) = 2$ and $rank(\mathscr{C}_2) = 1$, whereas the sets E_1^* and E_2^* are single lines as depicted in Fig. 1, in fact these points $\boldsymbol{m}$ are the solutions of the linear system

$$0 = \begin{bmatrix} \boldsymbol{C\Lambda\Pi}_i & \boldsymbol{C}_c \end{bmatrix} \begin{bmatrix} \boldsymbol{m} \\ u \end{bmatrix} \tag{9}$$

In the configuration $\mathcal{C}_1$, the set of reachable markings from an equilibrium one is the shadowed area that includes E_1^*, thus there must be two independent vectors describing this area, hence $rank(\mathscr{C}_1) = 2$. Since this area includes E_1^*, this line is one of the independent vectors. The other one could be taken as one orthogonal to E_1^*. However, since the marking in places p_3, p_1 is not infinite (it is bounded by the P-semiflows) and the control actions must be suitably bounded, then only the shadow area is reachable. In the configuration, $\mathcal{C}_2$, the set of reachable markings from an equilibrium one is the line of equilibrium markings E_2^*. Roughly speaking, the rank of the controllability matrices explains the dimension of the minimal affine varieties that include the reachability sets.

Notice that, even if $rank(\mathscr{C}_2) = 1$, the system is controllable over E_2^*, i.e., the rank-controllability does not characterize the controllability over equilibrium markings. However, rank-controllability, under some assumptions,[2] is a sufficient condition for controllability over the equilibrium markings, moreover, it depends only on the net structure and timing, and thus it is easy to verify.

Finally, let us extend the concept of rank-controllability and rank-observability from configurations to complete nets.

Definition 13. Consider a timed $TCPN$ structure $\langle \mathcal{N}, \lambda \rangle$. If all the configurations of the timed structure are rank-controllable (resp. rank-observable), then the $TCPN$ is said to be rank-controllable (resp. rank-observable).

Now, considering the concept of duality in linear systems (LS), where a LS is controllable iff its dual is observable, this work extends this concept to $TCPN$. Translating it to $TCPN$, however, is not a straightforward task since the duality concept introduced in Definition 2 is insufficient to study the relationship between rank-controllability and rank-observability. In particular, the dual operation does not invert the token/information flow determined by the PN structure, and this flow information determines the controllability and observability (see [18,22]). This fact will be explained using the net depicted in Fig. 2.(a). If the flow of the join transition t_a is equal to zero because the marking in p_1 (an input place of t_a) is equal to zero, then the marking variations of other input places (for instance p_2) to t_a are stopped at t_a, i.e. these variations are not carried out to downstream places of t_a. Nevertheless, in the dual computed as in Definition 2 (see Fig. 2.(b)), the marking variations of p_a' are never stopped at t_1', leading to inconsistencies in subsequent analysis. Moreover, the net of Fig. 2.(a) has two configurations (i.e. its marking evolution is represented by two LS) while its dual has only one. In order to overcome these issues, this work will define the dual for some subclasses of $TCPN$ and will derive the relation between the rank-controllability and rank-observability of the primal and dual respectively.

The following sections are devoted to show the relationship between rank-controllability and rank-observability through net's duality.

[2] Rank-controllability may not be sufficient for controllability in general nets. For instance, in non-live systems: due to the boundedness of the input, even if the system is rank-controllable, it does not guarantee that any existing deadlock can be avoided. Therefore, in non-live systems, rank-controllability is not sufficient nor necessary [17].

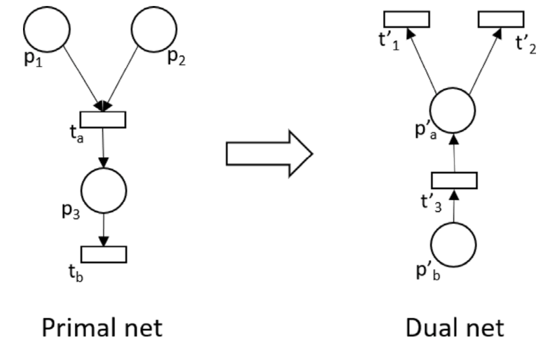

Fig. 2. A Primal net (left) and its dual net (right).

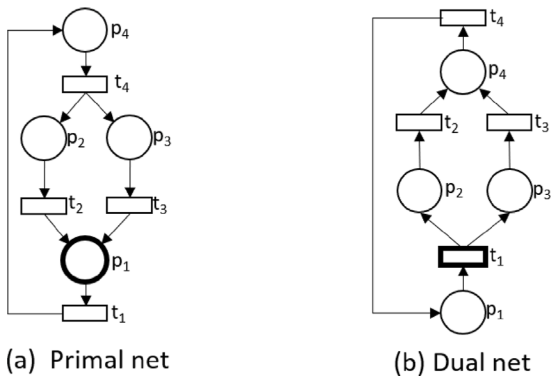

Fig. 3. (a) A FA net and (b) its dual.

4. Duality in Fork-Attribution nets

As already mentioned, this subclass is closed under the dual operation of Definition 2. In the sequel, an upper-index p (resp. d) will be used to denote matrices net structure and marking related to the primal (resp. dual) TCPN system. Any FA net has only one configuration, thus the marking evolution is described by a linear system. The next procedure computes the dual of a FA net.

Procedure 1. *Duality in Fork-Attribution TCPN.*

Input: A Fork-Attribution $TCPN$ structure $\langle \mathcal{N}^p, \lambda^p \rangle$ named the primal net. Without loss of generality, suppose that the transition t_i is being constrained by place p_i, for each $t_i \in T$ in the primal.
Output: The dual $TCPN$ structure, denoted as $\langle \mathcal{N}^d, \lambda^d \rangle$.

1: *Obtain the structure of the dual net $\mathcal{N}^d$ following Definition 2, i.e. places and transitions are interchanged and arcs are reversed.*
2: *Define the transition rates as equal, i.e. $\lambda^d = \lambda^p$*
3: *Define the set of controllable transitions such that t_i is controllable in the dual iff p_i is measurable in the primal. Similarly, p_i is measurable in the dual iff t_i is controllable in the primal.*

Since Procedure 1 only involves structural transformations and simple set operations, its complexity is linear on the number of nodes of the net.

This section focuses on strongly connected and bounded FAs, their dynamics are represented by a single positive linear system where the marking is bounded due to the only P-semiflow (see Proposition 3). Moreover, if the transitions are numbered as indicated in Procedure 1, then $\boldsymbol{\Pi}^p = \boldsymbol{\Pi}^d$ and they are diagonal matrices. Notice that the dual of the dual net is the primal.

Example 2. Consider the FA of Fig. 3(a) with $\lambda^p = [\lambda(t_1), \lambda(t_2), \lambda(t_3), \lambda(t_4)]$. Using Procedure 1, its dual $TCPN$ is computed and it is showed in Fig. 3(b), where $\lambda^d = \lambda^p$. Notice that the fork transition t_4 in the primal net is transformed into the attribution place p_4' in the dual. It is clear that the dual is also a FA, and that their incidence matrices are related as $\boldsymbol{C}^d = (\boldsymbol{C}^p)^T$.

4.1. Relations between observability and controllability based on duality for Fork-Attribution nets

The controllability and observability matrices of a primal and its dual are related in a precise manner in FA. This will be formally stated in the forthcoming Theorem 1. The following lemma presents how the dual matrices, such as the incidence, the input, the output and dynamic matrices are related to their respective matrices in the primal.

Lemma 1. *Let $\langle \mathcal{N}^p, \lambda^p \rangle$ be a Fork-Attribution $TCPN$ timed structure where T_c is the set of controllable transitions and P_O is the set of measurable places. Let $\langle \mathcal{N}^d, \lambda^d \rangle$ be its dual. Then the following relationships are obtained:*

Net matrices	*Primal*	*Dual*
Incidence	$\boldsymbol{C}^p$	$\boldsymbol{C}^d = (\boldsymbol{C}^p)^T$
Configuration	$\boldsymbol{\Pi}^p$	$\boldsymbol{\Pi}^d = \boldsymbol{\Pi}^p$
Input	$\boldsymbol{C}^p\boldsymbol{S}$	$\boldsymbol{C}^d\boldsymbol{O}^{pT}$
Output	$\boldsymbol{O}^p$	$\boldsymbol{O}^d = \boldsymbol{S}^T$
Dynamics	$\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p$	$\boldsymbol{C}^d\boldsymbol{\Lambda}^d\boldsymbol{\Pi}^d = (\boldsymbol{C}^p)^T\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p$

Theorem 1. *Let $\langle \mathcal{N}^p, \lambda^p \rangle$ and $\langle \mathcal{N}^d, \lambda^d \rangle$ be the primal and its dual Fork-Attribution timed $TCPN$ structures. Then, the following relations hold between the observability matrix and the controllability matrix of the primal $TCPN$ and its dual:*

$$\begin{bmatrix} \mathcal{O}^p \\ \boldsymbol{O}^p(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|} \end{bmatrix}^T = \begin{bmatrix} \boldsymbol{O}^{pT}, \boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p\mathscr{C}^d \end{bmatrix} \tag{10}$$

$$\begin{bmatrix} \mathcal{O}^d \\ \boldsymbol{O}^d(\boldsymbol{C}^d\boldsymbol{\Lambda}^d\boldsymbol{\Pi}^d)^{|P|} \end{bmatrix}^T = \begin{bmatrix} \boldsymbol{O}^{dT}, \boldsymbol{\Pi}^d\boldsymbol{\Lambda}^d\mathscr{C}^p \end{bmatrix} \tag{11}$$

$$\mathscr{C}^d = \boldsymbol{C}^d \cdot (\mathcal{O}^p)^T \tag{12}$$

$$\mathscr{C}^p = \boldsymbol{C}^p \cdot (\mathcal{O}^d)^T. \tag{13}$$

Proof. Let us firstly demonstrate (10). Recalling that the observability matrix of the primal net is

$$\mathcal{O}^p = \begin{bmatrix} \boldsymbol{O}^p \\ \boldsymbol{O}^p \cdot \boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p \\ \vdots \\ \boldsymbol{O}^p \cdot (\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|-1} \end{bmatrix},$$

then

$$\begin{bmatrix} \mathcal{O}^p \\ \boldsymbol{O}^p(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|} \end{bmatrix}^T = \begin{bmatrix} \boldsymbol{O}^{pT} & (\boldsymbol{O}^p\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^T & \dots & (\boldsymbol{O}^p(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|})^T \end{bmatrix} =$$

$$= \begin{bmatrix} \boldsymbol{O}^{pT} & \boldsymbol{\Pi}^{pT}\boldsymbol{\Lambda}^{pT}\boldsymbol{C}^{pT}\boldsymbol{O}^{pT} & \dots & (\boldsymbol{\Pi}^{pT}\boldsymbol{\Lambda}^{pT}\boldsymbol{C}^{pT})^{|P|}\boldsymbol{O}^{pT} \end{bmatrix} \tag{14}$$

Since $\boldsymbol{\Lambda}^p$ and $\boldsymbol{\Pi}^p$ are diagonal matrices in FA, then $\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p = \boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p$ and $(\boldsymbol{\Pi}^{pT}\boldsymbol{\Lambda}^{pT}\boldsymbol{C}^{pT})^k = (\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p\boldsymbol{C}^{pT})^k = \boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p(\boldsymbol{C}^{pT}\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{k-1}\boldsymbol{C}^{pT} = \boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p(\boldsymbol{C}^{pT}\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{k-1}\boldsymbol{C}^{pT}$, for any $k > 1$. Thus, (14) can be expressed as

$$\begin{bmatrix} \mathcal{O}^p \\ \boldsymbol{O}^p(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|} \end{bmatrix}^T = \begin{bmatrix} \boldsymbol{O}^{pT} & \boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p\boldsymbol{C}^{pT}\boldsymbol{O}^{pT} & \dots & \boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p(\boldsymbol{C}^{pT}\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|-1}\boldsymbol{C}^{pT}\boldsymbol{O}^{pT} \end{bmatrix} \tag{15}$$

Since $\mathscr{C}^d = \begin{bmatrix} \boldsymbol{C}^d\boldsymbol{O}^{pT}, (\boldsymbol{C}^d\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)\boldsymbol{C}^d\boldsymbol{O}^{pT}, \dots, (\boldsymbol{C}^d\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|-1}\boldsymbol{C}^d\boldsymbol{O}^{pT} \end{bmatrix}$ and $\boldsymbol{C}^d = \boldsymbol{C}^{pT}$, Eq. (15) is equivalent to (10). The relation (11) can be proved in a similar way.

To prove (12), consider the first entries of the matrices in (15) with the equality $\boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p = \boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p$, i.e. $\mathcal{O}^{pT} = \begin{bmatrix} \boldsymbol{O}^{pT} & \boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p\boldsymbol{C}^{pT}\boldsymbol{O}^{pT} & \dots & \boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p(\boldsymbol{C}^{pT}\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|-2}\boldsymbol{C}^{pT}\boldsymbol{O}^{pT} \end{bmatrix}$. By Lemma 1 $\boldsymbol{C}^{pT} = \boldsymbol{C}^d$ and premultiplying the previous equation by $\boldsymbol{C}^d$ it follows

$$\boldsymbol{C}^d\mathcal{O}^{pT} = \begin{bmatrix} \boldsymbol{C}^d\boldsymbol{O}^{pT} & \boldsymbol{C}^d\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p\boldsymbol{C}^d\boldsymbol{O}^{pT} & \dots & (\boldsymbol{C}^d\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|-1}\boldsymbol{C}^d\boldsymbol{O}^{pT} \end{bmatrix}$$

The previous equation is equivalent to (12). The relation (13) can be demonstrated in a similar way. ■

Corollary 1. *Consider a Fork-Attribution $TCPN$ and its dual. Then,*

$$rank(\mathcal{O}^p) = rank(\begin{bmatrix} \boldsymbol{O}^{pT}, \mathscr{C}^d \end{bmatrix}) \tag{16}$$

Proof. First, consider (10) being multiplied on the left by $(\boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p)^{-1}$. Since $(\boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p)^{-1}$ is a full rank diagonal matrix, it follows that:

$$rank\left(\begin{bmatrix}\mathcal{O}^p\\ \boldsymbol{O}^p(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|}\end{bmatrix}\right)=rank(\left[\boldsymbol{O}^{pT},\boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p\mathscr{C}^d\right])=$$

$$=rank(\left[(\boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p)^{-1}\right]\left[\boldsymbol{O}^{pT},\boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p\mathscr{C}^d\right])=rank\left(\left[(\boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p)^{-1}\boldsymbol{O}^{pT},\mathscr{C}^d\right]\right)$$

By the Cayley–Hamilton theorem, the leftmost term of previous equation is equal to $rank(\mathcal{O}^p)$. Finally, since by definition $\boldsymbol{O}^{pT}$ is a matrix whose columns are elementary vectors, and $\boldsymbol{\Pi}^p\boldsymbol{\Lambda}^p$ is a full rank diagonal matrix, the right term of the previous equation is equal to $rank(\left[\boldsymbol{O}^{pT},\mathscr{C}^d\right])$ thus (16) follows. ■

Rank-controllability and rank-observability are dual properties for consistent and strongly connected FA nets (i.e., for conservative FA nets). This is formalized in the following proposition.

Proposition 6. *Let $\langle\mathcal{N}^p,\lambda^p\rangle$ and $\langle\mathcal{N}^d,\lambda^d\rangle$ be a primal net and its dual, respectively, timed $TCPN$ structure, where $\mathcal{N}^p$ is a C_t and C_v Fork-Attribution. Then, $\langle\mathcal{N}^p,\lambda^p\rangle$ is rank-controllable iff $\langle\mathcal{N}^d,\lambda^d\rangle$ is rank-observable.*

Proof. (Sufficiency) Let $\langle\mathcal{N}^d,\lambda^d\rangle$ be rank-observable. This implies that $\mathcal{O}^d$ (the observability matrix of the dual net) has rank $|P|$. On the other hand, by the Sylvester's rank inequality [25], $rank(\boldsymbol{C}^p\cdot\mathcal{O}^{dT})\geq rank(\boldsymbol{C}^p)+rank(\mathcal{O}^d)-|P|$. Because of rank-observability: $rank(\mathcal{O}^d)=|P|$, and the inequality becomes $rank(\boldsymbol{C}^p\cdot\mathcal{O}^{dT})\geq rank(\boldsymbol{C}^p)$. However, the rank of $\left(\boldsymbol{C}^p\cdot\mathcal{O}^{dT}\right)$ cannot be greater than $rank(\boldsymbol{C}^p)$, then $rank(\boldsymbol{C}^p\cdot\mathcal{O}^{dT})=rank(\boldsymbol{C}^p)$. Moreover (12) implies $rank(\boldsymbol{C}^p\cdot\mathcal{O}^{dT})=rank(\mathscr{C}^p)$. Therefore, $rank(\mathscr{C}^p)=rank(\boldsymbol{C}^p)$. Since the rank of $\boldsymbol{C}^p$ plus its nullity is equal to its dimension (i.e. $rank(\boldsymbol{C}^p)+rank(\boldsymbol{B}_y)=|P|$), then $rank(\mathscr{C}^p)=rank(\boldsymbol{C}^p)=|P|-rank(\boldsymbol{B}_y)$, which implies that $\langle\mathcal{N}^p,\lambda^p\rangle$ is rank-controllable.

(Necessity) Assume that $\langle\mathcal{N}^p,\lambda^p\rangle$ is rank-controllable; then $rank(\mathscr{C}^p)=|P|-rank(\boldsymbol{B}_y)$. Since $\mathcal{N}^p$ is a consistent and strongly connected Fork-Attribution, then it is conservative and $rank(\boldsymbol{B}_y)=1$ (Proposition 3), and there exists a vector $\mathbf{y}>\mathbf{0}$ such that $\mathbf{y}^T\boldsymbol{C}^p=\mathbf{0}$. Thus, $rank(\mathscr{C}^p)=|P|-1$. On the other hand, the controllability matrix can be represented as $\mathscr{C}^p=\left[\boldsymbol{C}^p\boldsymbol{S},(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)\boldsymbol{C}^p\boldsymbol{S},\dots,(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|-1}\boldsymbol{C}^p\boldsymbol{S}\right]$ where $\boldsymbol{S}$ is a selector matrix for the controllable transitions. Multiplying the last expression from the left by $\mathbf{y}^T$ it results $\mathbf{y}^T\mathscr{C}^p=\left[\mathbf{y}^T\boldsymbol{C}^p\boldsymbol{S},\mathbf{y}^T(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)\boldsymbol{C}^p\boldsymbol{S},\dots,\mathbf{y}^T(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p)^{|P|-1}\boldsymbol{C}^p\boldsymbol{S}\right]=\mathbf{0}$.

On the other hand, for any elementary vector $\mathbf{e}_i$ it holds $\mathbf{y}^T\mathbf{e}_i>0$ since all the elements of $\mathbf{y}^T$ are positive. Therefore $\mathbf{e}_i$ cannot be expressed as a linear combination of the columns of $\mathscr{C}^p$, i.e. $\nexists\mathbf{x}$ s.t. $\mathbf{e}_i=\mathscr{C}^p\mathbf{x}$ (otherwise $\mathbf{y}^T\mathscr{C}^p\mathbf{x}=\mathbf{y}^T\mathbf{e}_i$ would lead to a contradiction). Consequently, $rank([\mathbf{e}_i,\ \mathscr{C}^p])>rank(\mathscr{C}^p)$. Since $\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p$ is a diagonal matrix in Fork-Attribution $TCPN$, it follows $rank([\mathbf{e}_i,\ \boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p\mathscr{C}^p])=rank([\mathbf{e}_i,\ \mathscr{C}^p])>rank(\mathscr{C}^p)$. Furthermore, the columns of the dual's output matrix $\boldsymbol{O}^d$ are elementary vectors, and there is at least one sensor, then

$$rank([\boldsymbol{O}^d,\ \boldsymbol{\Lambda}^d\boldsymbol{\Pi}^d\mathscr{C}^p])=|P|$$

and by (11), $rank(\mathcal{O}^d)=|P|$, which implies $\langle\mathcal{N}^d,\lambda^d\rangle$ is rank-observable. ■

Example 3. Consider the primal timed $TCPN$ structure $\langle\mathcal{N}^p,\lambda^p\rangle$ depicted in Fig. 3.(a), with $\lambda^p=[1,1,1,1]$. Assume that p_1 is the only measurable place. This net is FA. The incidence matrix $\boldsymbol{C}$, output matrix $\mathbf{O}^p$ and rate matrix $\boldsymbol{\Lambda}$ are:

$$\boldsymbol{C}^p=\begin{bmatrix}-1&1&1&0\\0&-2&0&1\\0&0&-2&1\\1&0&0&-1\end{bmatrix},\quad \boldsymbol{O}^p=\begin{bmatrix}1&0&0&0\end{bmatrix},\quad \boldsymbol{\Lambda}=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix} \tag{17}$$

Then the primal observability matrix is:

$$\mathcal{O}^p=\begin{bmatrix}\boldsymbol{O}^p\\ \boldsymbol{O}^pC\Lambda\Pi\\ \vdots\\ \boldsymbol{O}^p(C\Lambda\Pi)^{|P|-1}\end{bmatrix}=\begin{bmatrix}1&0&0&0\\-1&1&1&0\\1&-3&-3&2\\1&7&7&-8\end{bmatrix} \tag{18}$$

The rank of the observability matrix is three (the second and third columns are identical) while the number of places is four, i.e. the net is not rank-observable. In fact the markings $m(p_1)$, $m(p_4)$ can be estimated and the addition $m(p_2)+m(p_3)$ can also be estimated, but neither $m(p_2)$ nor $m(p_3)$ can be independently estimated. The dual of this net is depicted in Fig. 3.(b). According to the transformation procedure, transition t_1 is controllable.

The incidence matrix of the dual net $\boldsymbol{C}^d$, its input matrix $\boldsymbol{C}^d\mathbf{O}^{pT}$ and its controllability matrix $\mathscr{C}^d$ are:

$$\boldsymbol{C}^d=\begin{bmatrix}-1&0&0&1\\1&-2&0&0\\1&0&-2&0\\0&1&1&-1\end{bmatrix},\quad \boldsymbol{C}^d\boldsymbol{O}^{pT}=\begin{bmatrix}-1\\1\\1\\0\end{bmatrix},\quad \mathscr{C}^d=\begin{bmatrix}-1&1&1&9\\1&-3&7&-13\\1&-3&7&-13\\0&2&-8&22\end{bmatrix} \tag{19}$$

The rank of the controllability matrix is two (the second and third rows are identical). Since the incidence matrix has rank equal to three, then the net is not rank-controllable. In fact, the marking in places p_2 and p_3 cannot be independently controlled.

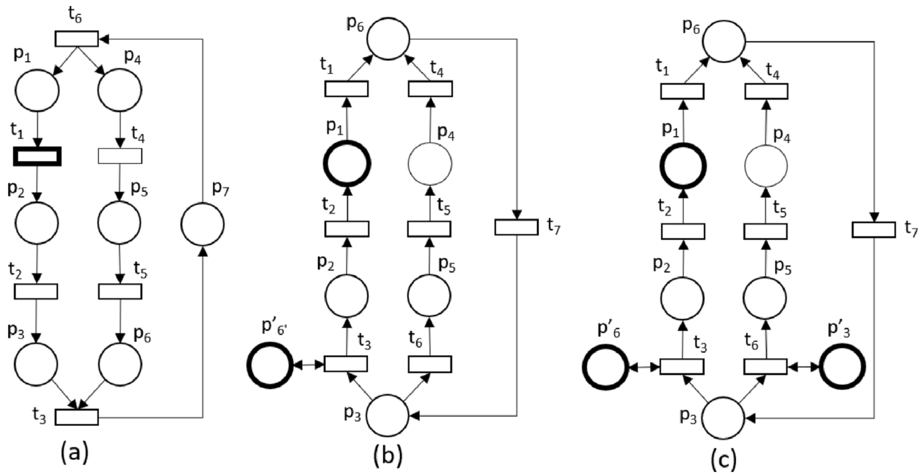

Fig. 4. (a) A Choice-Free $TCPN$. (b) The dual of configuration when p_3 constrains t_3. (c) The dual net of CF depicted in (a).

Example 4. Let us consider the same FA of Example 3, but the primal output matrix being redefined as:

$$\boldsymbol{O}^p = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{20}$$

It means that places p_1 and p_2 are measurable. In this case, the observability matrix has rank four, therefore the $TCPN$ is rank-observable. In the dual net, the input matrix changes to:

$$\boldsymbol{C}^d \boldsymbol{O}^{pT} = \begin{bmatrix} -1 & 0 \\ 1 & -2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{21}$$

The rank of the controllability matrix is equal to three. Since the rank of the incidence matrix is also three, then the net is rank-controllable, as expected.

5. Duality in Choice-free nets

The new element that a CF has in contrast to FA is the join, or rendez-vous, transition. Since these transitions have more than one input place, their enabling degree is affected by the min operator (see Eq. (1)), leading to the fact that the marking evolution is governed by different modes. Therefore, the use of Procedure 1 to compute the dual of a FA should be discarded because it is useful when there exists only one configuration. In order to compute the dual of a CF net, all the configurations should be considered. Unfortunately, enumerating all the configurations will lead to a computationally intractable problem. To overcome this inconvenience, this work presents a procedure to compute the dual of a CF net that does not need the enumeration of configurations. However, for the sake of explanation, this work mentions how to compute the dual of all the configurations of a CF net and then merge all of them to compute the dual of a CF net.

Let us give an intuition on the construction of dual nets for CF nets, by considering the net depicted in Fig. 4.(a). This net has two configurations, one when p_3 constrains the flow of t_3 and the other when p_6 constrains the flow of t_3. It is assumed that t_1 is controllable. Considering the configuration where p_6 is constraining the flow of t_3, the control action of t_1 is propagated downstream producing marking variations in p_3. Because of the configuration, however, this control action is blocked at p_3. The dual of this configuration is presented in Fig. 4.(b). It is computed using Procedure 1 where p_1 is measured and an extra self-loop place p'_6 is added. The marking of p'_6 is chosen in such a way that it is always constraining transition t_3 (i.e. in the primal p_6 is constraining t_3 then in the dual p'_6 is constraining t_3). Hence the backwards method can be used to compute the marking in p_2 in the dual, however, the marking of p_3 cannot be computed since this procedure is blocked at t_3 as required, i.e. in the primal the control action is blocked at t_3 and the marking observation is blocked at p_3. Fig. 4.(c) depicts the dual net of the primal net shown in Fig. 4.(a). The primal has two configurations, thus the dual must have two *valid* configurations. The valid configurations are when place p'_6 or p'_3 are constraining their respective output transitions.

The following procedure computes the dual of a CF net.

Procedure 2. *Dual-configuration and dual of a CF.*

Input: A CF timed structure $\langle \mathcal{N}^p, \lambda^p \rangle$.
Output: The dual TCPN denoted as $\langle \mathcal{N}^d, \lambda^d \rangle$.

1: If a transition t_j is controllable, then add a self-loop place p_{sj} to t_j that is always constraining the flow of t_j.
*2: **for** each valid configuration C_i of $\langle \mathcal{N}^p, \lambda^p \rangle$ **do***
3: Remove the arcs not belonging to C_i. Relabel transitions such that p_k is constraining transition t_k.
4: Compute an (almost) dual net using Procedure 1.
5: Add a measurable self-loop place around transitions whose respective places are not constraining the transition flow in the primal net. At this point, the derived net is named the related configuration C'_i of configuration C_i. Adding the reverse of removed arcs the dual $\mathcal{N}^d_i$ of the configuration C_i is derived.
*6: **end for***
7: Compute the dual net $\mathcal{N}^d$ as the union of all $\mathcal{N}^d_i$. The rate vector is the same $\lambda^p = \lambda^d_i$. In the dual net t_j, p_k are transformed from p_j, t_k of the primal, respectively. Added self-loop places are numbered consecutively to the rest of the places.

It is important to note that while the procedure is initially defined over the configurations of the primal, this is primarily done to illustrate the reasoning behind constructing the dual in this manner. In practice, however, the dual of a net can be obtained without the requirement of explicitly enumerating all the configurations of the primal. It can be computed by only replacing transitions by places, places by transitions, reversing the arcs, and adding self-loop places to the transitions in conflict. Hence, the complexity of the algorithm is linear on the number of nodes of the net. During this work, the following definition will be used.

Definition 14. Let C_i be a configuration of $\langle \mathcal{N}^p, \lambda^p \rangle$. The related configuration of C_i is the dual configuration C'_i computed by Procedure 2 at step 5.

5.1. Relations between observability and controllability based on duality for Choice-Free nets

The following lemma relates the dynamic matrices of primal and dual CF nets in a precise manner. The added self-loop places must be considered since they introduce null rows in the dynamic matrix, increasing its size.

Lemma 2. *Let $\langle \mathcal{N}^p, \lambda^p \rangle$ be a CF timed structure, where $\boldsymbol{S}$ is the selector matrix of controllable transitions, and $\boldsymbol{O}^p$ is the selector matrix of measurable places. Let $\langle \mathcal{N}^d, \lambda^d \rangle$ be its dual. Then the following relationships are valid (in the table the symbol $*$ means that this value is different from zero, but its value is not relevant for the present work).*

Net Matrices	*Primal*	*Dual*
Incidence	$\boldsymbol{C}^p$	$\boldsymbol{C}^d = \begin{bmatrix} (C^p)^T \\ \boldsymbol{0} \end{bmatrix}$
Configuration	$\boldsymbol{\Pi}^p_i$	$\boldsymbol{\Pi}^d_i = \begin{bmatrix} (\boldsymbol{\Pi}^p_i)^T & * \end{bmatrix}$
Input	$\boldsymbol{C}^p_c = \boldsymbol{C}^p\boldsymbol{S}$	$\boldsymbol{C}^d\boldsymbol{O}^{pT}$
Output	$\boldsymbol{O}^p$	$\boldsymbol{O}^d = \begin{bmatrix} \boldsymbol{S}^T & \boldsymbol{0} \end{bmatrix}$
Dynamic	$\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p_i$	$\boldsymbol{C}^d\boldsymbol{\Lambda}^d\boldsymbol{\Pi}^d_i = \begin{bmatrix} (C^p)^T\boldsymbol{\Lambda}^d(\boldsymbol{\Pi}^p_i)^T & * \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}$

Proof. Let $\mathcal{N}^p$ be a CF primal net. According to Procedure 2, in its dual the places and transitions are interchanged, and the direction of arcs are reversed. Moreover, some self-loop places are added. Then the incidence matrix of the dual is the transpose of the incidence matrix of the primal, with some extra zero rows representing the self-loop places added. In a particular primal configuration matrix $\boldsymbol{\Pi}^p_i$, a place p_j is constraining a join transition t_k; in the dual, the respective added self-loop place p'_j is constraining any output transition t_h of the conflict associated to p_k. Thus $\frac{1}{\boldsymbol{Pre}(p'_j,t_h)}$ should appear in the dual configuration matrix. Nevertheless, since this value will be irrelevant in further analyses, the dual configuration matrix is formed by the transpose of the original configuration matrix and the values due to added self-loop places (indicated as *). The dual input matrix corresponds to the measurable places in the primal, similar to the output matrix, where extra zeros come from added self-loop places. Finally, the dynamic matrices follow their definition in $TCPN$ systems. ■

The following result relates the controllability matrix of the primal net, for a configuration C_i, and the observability matrix of its dual.

Proposition 7. *Let $\langle \mathcal{N}^p, \lambda^p \rangle$ be a CF timed structure and let $\langle \mathcal{N}^d, \lambda^d \rangle$ be its dual. Consider a configuration C_i of $\langle \mathcal{N}^p, \lambda^p \rangle$ and its related configuration C'_i of $\langle \mathcal{N}^d, \lambda^d \rangle$, then the following equation holds*

$$\begin{bmatrix} \mathscr{C}^p_i & \mathcal{M} \end{bmatrix} = (\boldsymbol{C}^d)^T \cdot (\mathscr{O}^d_i)^T \tag{22}$$

where $\mathcal{M}$ is defined as

$$\mathcal{M} = \left[(\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p_i)^{|P^p|}\boldsymbol{C}^p\boldsymbol{S}, \ldots, (\boldsymbol{C}^p\boldsymbol{\Lambda}^p\boldsymbol{\Pi}^p_i)^{|P^d|-1}\boldsymbol{C}^p\boldsymbol{S} \right], \tag{23}$$

i.e. it continues the expansion of the controllability matrix from the $|P^p|th$ term to the $|P^d|-1th$ term, in this case $|P^p|$ and $|P^d|$ denote the number of places of the primal and the dual nets, respectively.

Proof. The controllability matrix for $\langle \mathcal{N}^p, \lambda^p \rangle$ in the configuration C_i is

$$\mathscr{C}_i^p = \left[\boldsymbol{C}^p \boldsymbol{S}, (\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p) \boldsymbol{C}^p \boldsymbol{S}, \dots, (\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p)^{|P^p|-1} \boldsymbol{C}^p \boldsymbol{S} \right] \tag{24}$$

The observability matrix for $\langle \mathcal{N}^d, \lambda^d \rangle$ in configuration C_i' can be represented as

$$[\mathcal{O}_i^d]^T = \left[\boldsymbol{O}^{d^T}, (\boldsymbol{C}^d \boldsymbol{\Lambda}^d \boldsymbol{\Pi}_i^d)^T \boldsymbol{O}^{d^T}, \dots, (\boldsymbol{C}^d \boldsymbol{\Lambda}^d \boldsymbol{\Pi}_i^d)^{T^{|P^d|-1}} \boldsymbol{O}^{d^T} \right] \tag{25}$$

Using the last expression and Lemma 2, the right part of (22), $(\boldsymbol{C}^d)^T (\mathcal{O}_i^d)^T$, becomes:

$$\begin{bmatrix} \boldsymbol{C}^p & \boldsymbol{0} \end{bmatrix} \left[\begin{bmatrix} \boldsymbol{S} \\ \boldsymbol{0} \end{bmatrix}, \begin{bmatrix} ((\boldsymbol{C}^p)^T \boldsymbol{\Lambda}^d (\boldsymbol{\Pi}_i^p)^T)^T & \boldsymbol{0} \\ * & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{S} \\ \boldsymbol{0} \end{bmatrix}, \dots, \left(\begin{bmatrix} ((\boldsymbol{C}^p)^T \boldsymbol{\Lambda}^d (\boldsymbol{\Pi}_i^p)^T)^T & \boldsymbol{0} \\ * & \boldsymbol{0} \end{bmatrix} \right)^{|P^d|-1} \begin{bmatrix} \boldsymbol{S} \\ \boldsymbol{0} \end{bmatrix} \right] \tag{26}$$

And finally,

$$(\boldsymbol{C}^d)^T (\mathcal{O}_i^d)^T = \left[\boldsymbol{C}^p \boldsymbol{S} \quad \boldsymbol{C}^p (\boldsymbol{\Pi}_i^p \boldsymbol{\Lambda}^d \boldsymbol{C}^p) \boldsymbol{S}, \dots, \boldsymbol{C}^p (\boldsymbol{\Pi}_i^p \boldsymbol{\Lambda}^d \boldsymbol{C}^p)^{|P^d|-1} \boldsymbol{S} \right] \tag{27}$$

By the construction of the dual, for each place p_k in $\mathcal{N}^d$, $\forall t_j \in p_k^\bullet$, $\lambda^d(t_j) = \lambda^p(t_k)$, consequently $\boldsymbol{\Pi}_i^p \boldsymbol{\Lambda}^d = \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p$. Replacing this expression and reordering the terms in (27) the following expression is obtained:

$$(\boldsymbol{C}^d)^T (\mathcal{O}_i^d)^T = \left[\boldsymbol{C}^p \boldsymbol{S}, (\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p) \boldsymbol{C}^p \boldsymbol{S}, \dots, (\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p)^{|P^d|-1} \boldsymbol{C}^p \boldsymbol{S} \right] \tag{28}$$

From the last expression, Eq. (22) follows. ■

Let us assume that the CF net $\mathcal{N}$ is C_t and C_v. The following result relates the rank-observability and rank-controllability properties of $\mathcal{N}$ the primal and its dual nets.

Proposition 8. *Consider a C_t and C_v CF $TCPN$ with $\langle \mathcal{N}^p, \lambda^p \rangle$ where $\langle \mathcal{N}^d, \lambda^d \rangle$ is its dual net computed by Procedure 2, and the configuration C_i of $\langle \mathcal{N}^p, \lambda^p \rangle$ has a related configuration C_i'. Then the following relations hold:*

1. *If $\langle \mathcal{N}^d, \lambda^d \rangle$ is rank-observable then $\langle \mathcal{N}^p, \lambda^p \rangle$ is rank-controllable.*
2. *If $\langle \mathcal{N}^p, \lambda^p \rangle$ is rank-controllable and the self-loop places of $\langle \mathcal{N}^d, \lambda^d \rangle$ are measured, then $\langle \mathcal{N}^d, \lambda^d \rangle$ is rank-observable.*

Proof. Let us firstly prove the relation 1.

Let $\langle \mathcal{N}^d, \lambda^d \rangle$ be rank-observable in configuration C_i', i.e., $\mathcal{O}_i^d$ (the observability matrix of the dual net in the configuration C_i') has rank $|P^d|$. Let us apply the Sylvester's rank inequality to (22),

$$rank(\boldsymbol{C}^d \cdot (\mathcal{O}_i^d)^T) \geq rank(\boldsymbol{C}^d) + rank(\mathcal{O}_i^d) - |P^d|$$

Since $rank(\mathcal{O}_i^d) = |P^d|$, the inequality becomes $rank(\boldsymbol{C}^d \cdot (\mathcal{O}_i^d)^T) \geq rank(\boldsymbol{C}^d)$ However, the rank of $\left(\boldsymbol{C}^d \cdot (\mathcal{O}_i^d)^T\right)$ cannot be greater than $rank(\boldsymbol{C}^d)$, then $rank((\boldsymbol{C}^d)^T (\mathcal{O}_i^d)^T) = rank(\boldsymbol{C}^d)$. By using Proposition 7, it follows that $rank(\left[\mathscr{C}_i^p \ \mathcal{M}\right]) = rank((\boldsymbol{C}^d)^T (\mathcal{O}_i^d)^T) = rank(\boldsymbol{C}^d)$. By the Cayley–Hamilton theorem $rank(\left[\mathscr{C}_i^p \ \mathcal{M}\right]) = rank(\mathscr{C}_i^p)$, and then,

$$rank(\mathscr{C}_i^p) = rank(\boldsymbol{C}^d) = rank\left(\begin{bmatrix} \boldsymbol{C}^{p^T} \\ \boldsymbol{0} \end{bmatrix} \right) = rank(\boldsymbol{C}^p)$$

Then, $rank(\mathscr{C}_i^p) = rank(\boldsymbol{C}^p)$ implies that $\langle \mathcal{N}^p, \lambda^p \rangle$ in configuration C_i is rank-controllable. Since all the configurations in the dual, related to configurations in the primal, are rank-observable, then all the configurations in the primal are rank-controllable and the CF is rank-controllable.

Let us prove the relation 2.

Assume that $\langle \mathcal{N}^p, \lambda^p \rangle$ is rank-controllable, i.e. $rank(\mathscr{C}_i^p) = rank(\boldsymbol{C}^p)$. Since $\mathcal{N}^p$ is a strongly connected and C_t CF, then it is conservative (Proposition 2). Thus a basis $\boldsymbol{B}_y^T$ for the left annulers of $\boldsymbol{C}^p$ can be represented with nonnegative entries, i.e., each row $\mathbf{y}^T$ of $\boldsymbol{B}_y^T$ is a P-semiflow $\mathbf{y} \geq \mathbf{0}$. Then $\boldsymbol{B}_y^T$ is also a basis for the left annulers of $\mathscr{C}_i$. Let k be the number of rows of $\boldsymbol{B}_y^T$. Next, consider the first P-semiflow $\mathbf{y}_1^T$ of the basis $\boldsymbol{B}_y^T$ and define a row elementary vector $\mathbf{e}_j$ (i.e., a null vector excepting its jth entry which is one), where p_j is any place in $\|\mathbf{y}_1\|$. Then, $\mathbf{y}_1^T \mathbf{e}_j^T > 0$ but $\mathbf{y}_1^T \mathscr{C}_i = \mathbf{0}$, thus $rank([\mathbf{e}_j^T, \mathscr{C}_i]) > rank(\mathscr{C}_i)$. Therefore, if a set of places $\{p_{j_1}, \dots, p_{j_k}\}$ is selected such that each p_{j_r} belongs to $\|\mathbf{y}_r\|$, then it follows

$$rank([\mathbf{e}_{j_1}^T, \dots, \mathbf{e}_{j_k}^T, \mathscr{C}_i]) = |P| \tag{29}$$

Let $\langle \mathcal{N}_i^d, \lambda_i^d \rangle$ denote the dual FA structure obtained after the step 4. Since $\boldsymbol{\Pi}_i^d$ remains as diagonal and the number of places is equal to $|P| = |P_i^d|$, from (29) it follows that $rank([\mathbf{e}_{j_1}^T, \dots, \mathbf{e}_{j_k}^T, \boldsymbol{\Pi}^d \boldsymbol{\Lambda}^d \mathscr{C}_i]) = |P_i^d|$. Therefore, if the output matrix in the dual is such that $\boldsymbol{O}^{d^T} = [\mathbf{e}_{j_1}^T, \dots, \mathbf{e}_{j_k}^T, *]$, where each p_{j_r} is a place in the support of the rth P-semiflow of the primal, then by (22) of Proposition 7 it follows that $rank(\mathcal{O}_i^d) = |P_i^d|$, which implies that the dual $TCPN$ is rank-observable.

Finally, observability of the dual configuration nets imply observability of the dual, since this last net has the same structure that the dual configuration nets with the addition of self-loop places that are measurable by hypothesis.

To complete the proof, let us demonstrate that, under the given conditions, the output matrix of the dual net has the required form ${\boldsymbol{O}^{d}}^{T} = [\mathbf{e}_{j_1}^T, \ldots, \mathbf{e}_{j_k}^T, *]$. Proceeding by contradiction, suppose that for a given P-semiflow $\mathbf{y}_r^T$ in the primal, $\nexists p_{j_r} \in \|\mathbf{y}_r\|$ such that $\mathbf{e}_{j_r}$ is a row of $\boldsymbol{O}^d$. There are three possibilities:

(1) There is a place $p_{j_r} \in \|\mathbf{y}_r\|$ such that $(p_{j_r}, t_s) \notin C_i$. Since p_{j_r} is not constraining any transition, then in the dual it is transformed into transition t_{j_r} and a self-loop place p_{j_r} is also added in the dual. Since all the self-loop places are measurable in dual net then $\mathbf{e}_{j_r}$ appears in $\boldsymbol{O}^d$, a contradiction.

(2) There is a place $p_{j_r} \in \|\mathbf{y}_r\|$ such that $(p_{j_r}, t_s) \in C_i$, where t_s is the output transition of p_{j_r} and t_s is controllable. Thus, after the application of step 3 of Procedure 2, t_s and p_{j_r} share the same sub-index (since p_{j_r} constrains t_s and a relabeling is applied), i.e., t_s is t_{j_r}. The controllable transition t_{j_r} is translated into a measurable place p_{j_r} in the dual, thus $\mathbf{e}_{j_r}$ appears in $\boldsymbol{O}^d$, a contradiction.

(3) Each place $p_{j_r} \in \|\mathbf{y}_r\|$ constrains its output transition t_s, but it is uncontrollable. In such case, by Proposition 13 in the Appendix, the primal net is not rank-controllable, which is a contradiction. ■

Example 5. Consider the CF timed structure depicted in Fig. 4.a with a firing rate vector $\lambda = [1, 1, 1, 1, 1, 1]^T$. The dual net is depicted in Fig. 4.c. Considering $T_c = \{t_1\}$ in the primal, then the primal net is not rank-controllable (since the primal is not rank-controllable at the configuration in which p_6 constrains t_3). $T_c = \{t_1\}$ implies that p_1, p_3', p_6' are measurable in the dual. Therefore, it is not rank-observable at the configuration in which p_6' constrains t_3. Thus, the dual is not rank-observable, as stated in the previous theorem.

6. Duality in Join-Free nets

The analysis of JF nets is focused when all transitions in conflict are uncontrollable. In this case, the flows through the transitions in the conflict are proportional. Thus the Procedure 3 is used to remove conflicts. It allows obtaining an intermediate FA net (named reduced net) whose marking evolution has the same behavior as that of the original net. Afterwards, the FA results can be applied to the intermediate net.

Procedure 3. *Merging uncontrolled conflicts.*

Input: A JF timed structure $\langle \mathcal{N}, \lambda \rangle$.
Output: An equivalent $TCPN$ timed structure $\langle \mathcal{N}^R, \lambda^R \rangle$ without uncontrolled conflicts.

1: *Initialize the set of new transitions $T_{new} = \emptyset$ and the set of unchanged transitions $T_{unchanged} = T$.*
2: ***for** each place p_j constraining an uncontrolled conflict (i.e. a place with a postset $T_{p_j} = p_j^\bullet$ such that $|T_{p_j}| > 1$ and $T_{p_j} \cap T_c = \emptyset$) **do***
3: *Compute the output flow of p_j when its marking is 1, i.e., $f_{p_j} = [1, \ldots, 1] \cdot \boldsymbol{\Lambda}\boldsymbol{\Pi}(\bullet, j)$*
4: *Compute the column of $\boldsymbol{C}^R$ related to the new transition t_{p_j}, obtained by merging those in $p_j^\bullet$, as $c_{p_j} = \frac{1}{f_{p_j}} \boldsymbol{C}\boldsymbol{\Lambda}\boldsymbol{\Pi}(\bullet, j)$*
5: *Compute the rate of the new merged transition t_{p_j} as $\lambda_{p_j} = |\boldsymbol{C}(j, \bullet)\boldsymbol{\Lambda}\boldsymbol{\Pi}(\bullet, j)|$.*
6: *Update $T_{new} = T_{new} \cup \{t_{p_j}\}$ and $T_{unchanged} = T_{unchanged} - p_j^\bullet$.*
7: ***end for***
8: *$\boldsymbol{C}^R$ is built with the column vectors c_{p_j} defined for each new transition $t_{p_j} \in T_{new}$ and the columns of $\boldsymbol{C}$ related to the unchanged transitions $T_{unchanged}$.*
9: *λ^R is defined by the rates λ_{p_j} defined for each new transition $t_{p_j} \in T_{new}$ and the rates of the unchanged transitions $T_{unchanged}$.*

The complexity of Procedure 3 is polynomial on the number of places of the net. The for loop is executed at most $|P|$ times. Then, for each choice place of the net, lines 3 to 6 perform simple operations (including vector multiplications that contribute to this polynomial complexity), related to that place and its output transitions. Finally, lines 8 and 9 indicate how the equivalent timed structure is constructed.

According to this procedure, matrix $\boldsymbol{\Pi}^R$ of the resulting reduced net has the rows of $\boldsymbol{\Pi}$ for the unchanged transitions (i.e the same place continues limiting the transition), while the others are elementary and linearly independent (since every place is constraining a different transition), thus $\boldsymbol{\Pi}^R$ has full row rank. By construction, the dynamic matrices of $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ and $\langle \mathcal{N}^R, \lambda^R, \boldsymbol{m}_0 \rangle$, $\boldsymbol{C}\boldsymbol{\Lambda}\boldsymbol{\Pi} = \boldsymbol{C}^R\boldsymbol{\Lambda}^R\boldsymbol{\Pi}^R$, are identical, resulting on a FA system that evolves exactly as the original JF one [20,26]. Based on this reasoning, the following proposition can be stated:

Proposition 9. *Let $\langle \mathcal{N}, \lambda \rangle$ be a JF timed structure where all transitions in conflicts are uncontrollable and $\langle \mathcal{N}^R, \lambda^R \rangle$ be its equivalent timed structure computed by Procedure 3. The marking evolution of the reduced system $\langle \mathcal{N}^R, \lambda^R, \boldsymbol{m}_0 \rangle$ is equal to the marking evolution of the original one $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ for any given $\boldsymbol{m}_0 > 0$ and any bounded control action $\boldsymbol{u}(\tau)$.*

Proof. Procedure 3 is focused on merging uncontrolled conflicts, i.e. if there exists a place p_j with more than one output transition, then line 3 of the procedure computes the output flow of p_j by summation all the output flows of p_j. Line 4 computes the column of the incidence matrix of a new transition t_{p_j} that produce a normalized output flow in p_j, finally line 5 assigns the firing rate to t_{p_j} in such a way that the flow produced by t_{p_j} is equivalent to the flow produced by all output transitions of p_j. Thus, as mentioned above, the dynamic matrices of $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ and $\langle \mathcal{N}^R, \lambda^R, \boldsymbol{m}_0 \rangle$, $\boldsymbol{C}\boldsymbol{\Lambda}\boldsymbol{\Pi} = \boldsymbol{C}^R\boldsymbol{\Lambda}^R\boldsymbol{\Pi}^R$, are identical. The control is applied to transitions that are not in conflict, i.e. they are preserved identical in both nets, i.e. $\boldsymbol{C}^R\boldsymbol{u}^R(\tau) = \boldsymbol{C}\boldsymbol{u}(\tau)$. Hence $\dot{\boldsymbol{m}} = \boldsymbol{C}\boldsymbol{\Lambda}\boldsymbol{\Pi}_i\boldsymbol{m} - \boldsymbol{C}\mathbf{u} = \boldsymbol{C}^R\boldsymbol{\Lambda}^R\boldsymbol{\Pi}_i^R\boldsymbol{m} - \boldsymbol{C}^R\mathbf{u}^R$. ■

Moreover, since both nets $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ and $\langle \mathcal{N}^R, \lambda^R, \boldsymbol{m}_0 \rangle$ have the same marking evolution, and $\langle \mathcal{N}^R, \lambda^R \rangle$ is a FA net, then the results stated in Section 4 apply to JF nets. In particular, Proposition 6 allows us to state the following proposition.

Proposition 10. *Let $\langle \mathcal{N}, \lambda \rangle$ be a JF timed structure and $\langle \mathcal{N}^R, \lambda^R \rangle$ be its equivalent timed structure computed by Procedure 3. $\langle \mathcal{N}, \lambda, \boldsymbol{m}_0 \rangle$ is rank-controllable iff the dual of $\langle \mathcal{N}^R, \lambda^R, \boldsymbol{m}_0 \rangle$ is rank-observable.*

Proof. By applying Proposition 9, it can be noticed that $\langle \mathcal{N}, \lambda \rangle$ and $\langle \mathcal{N}^R, \lambda^R \rangle$ exhibit identical marking evolutions. Moreover, by using Procedure 3, the conflicts in $\langle \mathcal{N}, \lambda \rangle$ are merged in fork transitions in $\langle \mathcal{N}^R, \lambda^R \rangle$. Thus, the reduced net is a FA net. Then, using Proposition 6 the result follows. ■

7. Duality in proportional equal conflict nets

The results of this section are a consequence of that presented in previous sections. The proof for the case where no controllable transitions are in conflict is sketched based on previous results.

Procedure 4. *Dual nets for a Proportional Equal-Conflict Net.*

Input: A Proportional Equal-Conflict Net $TCPN$ $\langle \mathcal{N}^p, \lambda^p \rangle$.
Output: The dual $TCPN$ timed structure $\langle \mathcal{N}^d, \lambda^d \rangle$.

1: *Merge the transitions in conflict in $\langle \mathcal{N}^p, \lambda^p \rangle$ by applying Procedure 3. The resulting net is a CF net denoted $\langle \mathcal{N}^{cf}, \lambda^{cf} \rangle$.*
2: *Apply Procedure 2 to $\langle \mathcal{N}^{cf}, \lambda^{cf} \rangle$ in order to obtain the dual net $\langle \mathcal{N}^d, \lambda^d \rangle$ of $\langle \mathcal{N}^p, \lambda^p \rangle$.*

Notice that Procedure 4, which combines Procedure 1, Procedure 2, and Procedure 3, has a polynomial complexity due to the polynomial complexity of Procedure 3.

7.1. Relations between observability and controllability based on duality for Equal-Conflict nets

The following proposition establishes the relation between the controllability matrix of the primal and the observability matrix of the dual, as a generalization of Proposition 7.

Proposition 11. *Let $\langle \mathcal{N}^p, \lambda^p \rangle$ a PEQ timed structure, and let T_c be the set of controllable transitions of $\langle \mathcal{N}^p, \lambda^p \rangle$ such that no transition in T_c is in conflict ($\forall t_k \in T_c$, $({}^\bullet t_k)^\bullet = \{t_k\}$). Let $\langle \mathcal{N}^d, \lambda^d \rangle$ be its dual net as defined in Procedure 4. Consider a configuration C_i of $\langle \mathcal{N}^p, \lambda^p \rangle$ and its related configuration C_i' of $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$, then the following equation holds*

$$\left[\mathscr{C}_i^p \ \mathcal{M}\right] = (\boldsymbol{C}^{ds})^T \cdot (\mathscr{O}_i^{ds})^T \tag{30}$$

where $\mathcal{M}$ defined as

$$\mathcal{M} = \left[(\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p)^{|P^p|} \boldsymbol{C}^p \boldsymbol{S}, \ldots, (\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p)^{|P^{ds}|-1} \boldsymbol{C}^p \boldsymbol{S}\right], \tag{31}$$

i.e., it continues the expansion of the controllability matrix.

Proof. Step 3 of Procedure 3 merges transitions in conflict. This step changes the incidence, rate and configuration matrices, nevertheless the product $\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p$ remains unchanged. Similarly, the input matrix remains unchanged since no transition in T_c is merged. Consequently, the controllability matrix remains equal. Notice that, after merging the transitions in conflict, the resulting net corresponds to a Choice-Free $TCPN$. Then, by Proposition 7, Eq. (30) follows. ■

The following proposition can be seen as a generalization of Theorem 1 to PEQ nets. In this case, the relation appears at the level of the span of the matrices.

Proposition 12. *Let $\langle \mathcal{N}^p, \lambda^p \rangle$ be a PEQ timed structure with at least one join transition, and let T_c be the set of controllable transitions of $\langle \mathcal{N}^p, \lambda^p \rangle$ such that no transition in T_c is in conflict ($\forall t_k \in T_c$, $({}^\bullet t_k)^\bullet = \{t_k\}$). Let $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$ be its dual obtained by using Procedure 4. Consider a configuration C_i of $\langle \mathcal{N}^p, \lambda^p \rangle$ and its related configuration C_i' of $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$. Then*

$$span\left\{\mathscr{O}_i^{ds^T}\right\} = span\left\{\left[\boldsymbol{O}^{ds^T}, \boldsymbol{\Pi}_i^{ds^T} \boldsymbol{\Lambda}^{ds} \mathscr{C}_i^p\right]\right\} \tag{32}$$

Proof. Consider the following expression for the observability matrix of the configuration C_i' in the dual composition:

$$\mathscr{O}_i^{ds^T} = \left[\boldsymbol{O}^{ds^T}, (\boldsymbol{C}^{ds} \boldsymbol{\Lambda}^{ds} \boldsymbol{\Pi}_i^{ds})^T \boldsymbol{O}^{ds^T}, \ldots, (\boldsymbol{C}^{ds} \boldsymbol{\Lambda}^{ds} \boldsymbol{\Pi}_i^{ds})^{T^{|P^{ds}|-1}} \boldsymbol{O}^{ds^T}\right] \tag{33}$$

Applying the transpose of the terms

$$\mathscr{O}_i^{ds^T} = \left[\boldsymbol{O}^{ds^T}, (\boldsymbol{\Pi}_i^{ds^T} \boldsymbol{\Lambda}^{ds} \boldsymbol{C}^{ds^T}) \boldsymbol{O}^{ds^T}, \dots, (\boldsymbol{\Pi}_i^{ds^T} \boldsymbol{\Lambda}^{ds} \boldsymbol{C}^{ds^T})^{|P^{ds}|-1} \boldsymbol{O}^{ds^T} \right] \tag{34}$$

Reordering the terms

$$\mathscr{O}_i^{ds^T} = \left[\boldsymbol{O}^{ds^T}, \boldsymbol{\Pi}_i^{ds^T} \boldsymbol{\Lambda}^{ds} \boldsymbol{C}^{ds^T} \boldsymbol{O}^{ds^T}, \dots, \boldsymbol{\Pi}_i^{ds^T} \boldsymbol{\Lambda}^{ds} \left(\boldsymbol{C}^{ds^T} \boldsymbol{\Pi}_i^{ds^T} \boldsymbol{\Lambda}^{ds} \right)^{|P^{ds}|-2} \boldsymbol{C}^{ds^T} \boldsymbol{O}^{ds^T} \right] \tag{35}$$

The input, output and dynamic matrices relationships of Lemma 2 also holds for $PEQ\ TCPN$s, in particular $\boldsymbol{C}^{ds^T} \boldsymbol{O}^{ds^T} = \boldsymbol{C}^p \boldsymbol{S}$ and $\boldsymbol{C}^{ds^T} \boldsymbol{\Pi}_i^{ds^T} = \boldsymbol{C}^p \boldsymbol{\Pi}_i^p$, thus

$$\mathscr{O}_i^{ds^T} = \left[\boldsymbol{O}^{ds^T}, \boldsymbol{\Pi}_i^{ds^T} \boldsymbol{\Lambda}^{ds} \boldsymbol{C}^p \boldsymbol{S}, \dots, \boldsymbol{\Pi}_i^{ds^T} \boldsymbol{\Lambda}^{ds} (\boldsymbol{C}^p \boldsymbol{\Pi}_i^p \boldsymbol{\Lambda}^{ds})^{|P^{ds}|-2} \boldsymbol{C}^p \boldsymbol{S} \right] \tag{36}$$

By using $\boldsymbol{\Pi}_i^p \boldsymbol{\Lambda}^{ds} = \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p$ as in the proof of Proposition 7,

$$\mathscr{O}_i^{ds^T} = \left[\boldsymbol{O}^{ds^T}, \boldsymbol{\Pi}_i^{ds^T} \boldsymbol{\Lambda}^{ds} \underbrace{\left[\boldsymbol{C}^p \boldsymbol{S}, \cdots, \left(\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p \right)^{|P^{ds}|-2} \boldsymbol{C}^p \boldsymbol{S} \right]}_{(*)} \right] \tag{37}$$

Notice that the term $(*)$ in previous equation is the primal's controllability matrix expansion up to the exponent $|P^{ds}| - 2$. Now, since there is at least one join in the primal net then $|P^{ds}| \geq |P^p| + 1$ and according to the Caley-Hamilton theorem, the last columns of $(*)$ are linear combinations of the first $|P^p|$ terms of $\mathscr{C}_i^p$. In other words, the term $(*)$ generates the same image subspace than $\mathscr{C}_i^p$. Therefore (32) holds. ■

The relation (32) is an extension of that proved in [22] as the former can be applied to the dual net, while the later must be applied to each dual configuration.

The following proposition generalizes Proposition 8. It relates the rank-observability and rank-controllability properties of the primal and dual nets, assuming the nets are strongly connected and consistent.

Theorem 2. *Consider a $TCPN$ timed structure $\langle \mathcal{N}^p, \lambda^p \rangle$ such that $\mathcal{N}^p$ is a strongly connected and consistent PEQ net. Let T_c be the set of controllable transitions and assume that no transition in T_c is in conflict, i.e., $\forall t_k \in T_c$, $({}^\bullet t_k)^\bullet = \{t_k\}$. Let $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$ be its dual net. Consider a configuration C_i of $\langle \mathcal{N}^p, \lambda^p \rangle$ and its related configuration C_i' of $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$. If there exists a vector $\boldsymbol{v} > \boldsymbol{0}$ such that $\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p \boldsymbol{v} = \boldsymbol{0}$, then the following relations hold:*

1. *If $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$ is rank-observable then $\langle \mathcal{N}^p, \lambda^p \rangle$ is rank-controllable.*
2. *If $\langle \mathcal{N}^p, \lambda^p \rangle$ is rank-controllable and the self-loop places of $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$ are measured, then $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$ is rank-observable.*

Proof. $\langle \mathcal{N}^p, \lambda^p \rangle$ is dynamically equivalent to the Choice-Free net $\langle \mathcal{N}^{cf}, \lambda^{cf} \rangle$ obtained by applying step 3 Procedure 3. Since no controllable transition is merged then the input matrices are equal, thus the controllability matrices are the same. By construction, $\mathcal{N}^{cf}$ is strongly connected. Moreover, the existence of $\mathbf{v} > \mathbf{0}$ such that $\boldsymbol{C}^p \boldsymbol{\Lambda}^p \boldsymbol{\Pi}_i^p \mathbf{v} = \mathbf{0}$ implies that $\mathcal{N}^{cf}$ is consistent (subsection 6.2 of [27]). Therefore, the relations 1 and 2 hold iff they hold for $\langle \mathcal{N}^{cf}, \lambda^{cf} \rangle$ instead of $\langle \mathcal{N}^p, \lambda^p \rangle$.

On the other hand, by Proposition 8, the relations 1 and 2 hold between the Choice-Free net $\langle \mathcal{N}^{cf}, \lambda^{cf} \rangle$ and its dual. Moreover, according to the Procedure 4, $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$ is also the dual of $\langle \mathcal{N}^{cf}, \lambda^{cf} \rangle$. Therefore, the relations 1 and 2 holds between $\langle \mathcal{N}^{cf}, \lambda^{cf} \rangle$ and $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$, consequently, they hold between $\langle \mathcal{N}^p, \lambda^p \rangle$ and $\langle \mathcal{N}^{ds}, \lambda^{ds} \rangle$, thus the proposition holds. ■

8. Illustrative example

Fig. 5 presents a Flexible Manufacturing System (FMS). It is composed of two production lines. Both lines have an assembly section, each one divided into two sections, one intermediate buffer and one output store. Section one of each line has an assembly station and a polishing surface stage; while section two has a finishing stage. The system operates as follows. First, two parts A, B are simultaneously fed into their respective assembly stations. Type A and type B parts can be assembled in two different arrangements in the assembly stations. These two parts are immediately assembled. Afterwards the surface of this assemble is polished. Polished assembles are stored in the intermediate buffer area. During the finishing stage, assembles are covered with an antirust polyester paint.

The $TCPN$ model of the FMS is depicted in Fig. 6.(a). Transition t_{15} indicates that both parts are retrieved from the input stores and fed into the assembly sections; transitions t_1, t_2 represent the assembly stations of section one and two, respectively. Transitions t_3 and t_4 are modeling the loading of assembles to polishing machines of section one or two, respectively. Place p_7 represents the intermediate buffer area. Transitions t_7, t_8 represent that the assembles are loaded into finishing stages of section one or two, respectively. Finally for the finishing stage, transitions t_9 and t_{10} represent that assembles are loaded into the painting machines of section one or two, respectively. Transitions t_{11} and t_{12} represent the painting and in t_{13} and t_{14} the machines are discharged. For simplicity, $\forall i,\ \lambda(t_i) = 1$.

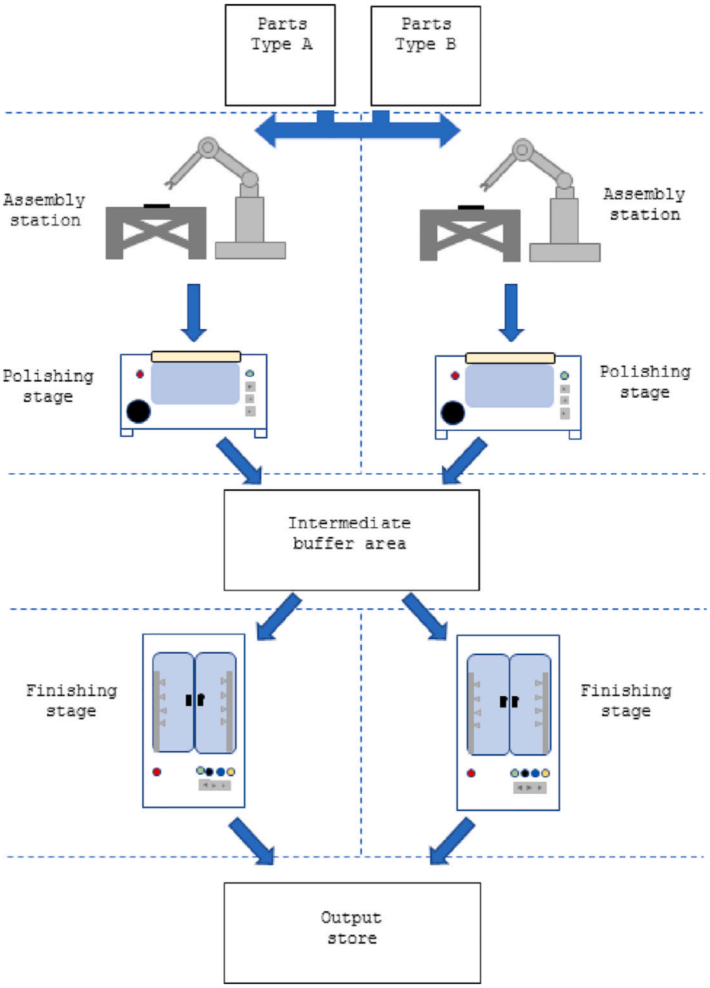

Fig. 5. Flexible manufacturing system composed of two identical production lines. Each line has an assembly station, a polishing stage, and a finishing stage.

Transitions t_{15}, t_3, t_4, t_9, t_{10}, t_{11}, t_{12} are controllable. Notice that the net has 64 configurations, i.e. all these configuration need to be analyzed to determine the rank-controllability of the net. Fortunately, following the results herein presented, the rank-controllability of the net can be stated in an easier way.

Since the net is PEQ, then its dual composition can be found with Procedure 4. The dual composition net is depicted in Fig. 6.(c). In this case p_{15}, p_3, p_4, p_9, p_{10}, p_{11}, p_{12} plus all self-loop places are measurable. According to the results presented in [15,18,19], it is easy to prove that this net is observable (thus rank-observable) for all the configurations. In detail, since all input places to join transitions are measurable, then the current configuration can be determined. Moreover, since place p_{15} is measurable then the marking of place p_1p_2 can be computed. Also places p_3, p_4 are measurable, and also their output flow is known, then the marking of places p_5, p_6 and p_7p_8 can be computed. Places p_9, p_{10}, p_{11} and p_{12} are measurable, and also their output flow is known, then the marking of p_{13} and p_{14} can be computed. According to Theorem 2, the primal net (i.e. the $TCPN$ modeling the system) is rank-controllable in all the configurations. It means that the throughput of the FMS can be controlled by slowing down the activities represented by transitions $t_3, t_4, t_9, t_{10}, t_{11}, t_{12}, t_{15}$.

Moreover, according to the results presented in [15,18,19], sensors in p_{11} and p_{12} in the dual composition net can be removed and the net is still rank-observable. Thus the control over transitions t_{11} and t_{12} can be removed without losing the controllability of the primal system.

9. Conclusions and future work

In this work, the relation between controllability and observability has been explored for some reported subclasses of $TCPN$s by using duality concepts. For this, the properties of rank-controllability and rank-observability have been used; the latter has been introduced in this work. Rank-controllability implies controllability over interior equilibrium points, provided $\boldsymbol{m}_0 > \mathbf{0}$, while rank-observability is equivalent to observability of the $TCPN$ in the corresponding configuration.

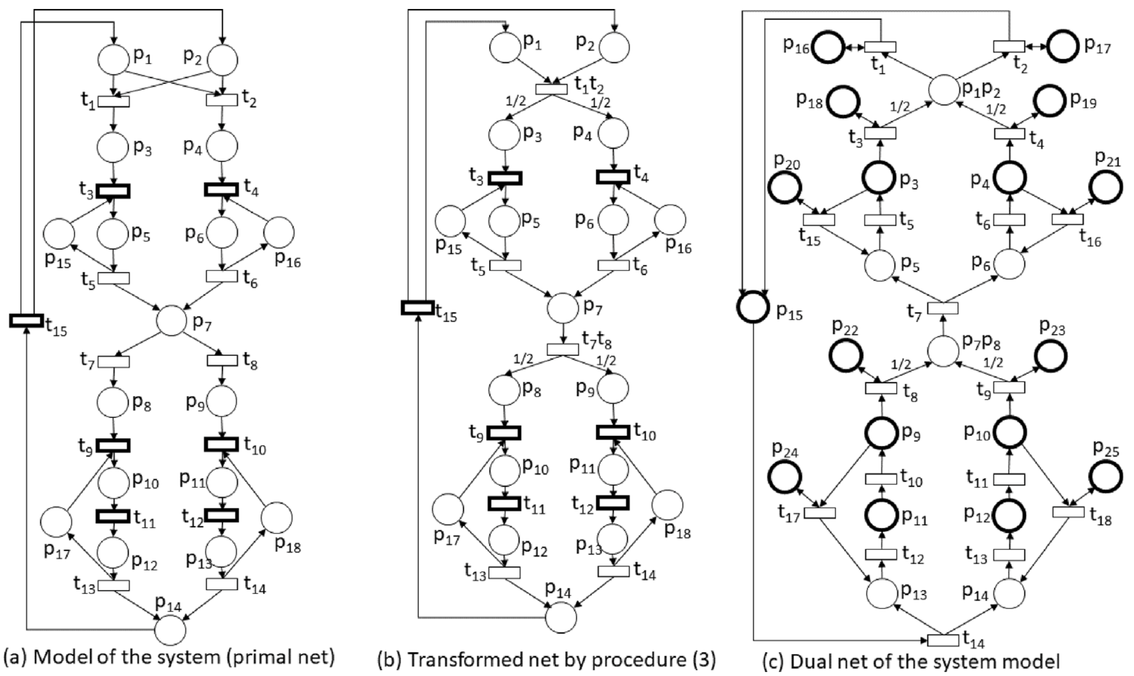

Fig. 6. (a) It represents the $TCPN$ model of the FMS described in the example of Section 8. Notice that it is a PEQ $TCPN$ and it is named the primal net. (b) It is the transformed net by Procedure 3. (c) It is the dual net of the primal net computed by Procedure 4.

Procedures have been proposed to compute the dual net of a given primal one, for the FA, CF, JF, and PEQ subclasses. Special attention was paid to elements generating non-linearities in the original and dual nets, such as join transitions, that generate several configurations, and choice places, that generate join transitions in the dual. Dual nets are computed for each possible configuration at the primal; in PEQ nets, a particular single dual net, named dual composition, was suitable for the properties analyzed in this work.

Based on the dual constructions, relations between the rank-observability and the rank-controllability properties have been investigated. In particular, it has been proved that a primal FA is rank-observable (resp. rank-controllable) iff its dual is rank-controllable (resp. rank-observable). This analysis was extended to CF, JF, and PEQ nets, however, in these subclasses the relations are non-symmetric. In particular, if the dual net is rank-observable then the primal is rank-controllable. The opposite implication holds only if, additionally, all the self-loop places in the dual are measurable.

In the future, the relations here presented will be used to derive structural conditions for controllability. Furthermore, by combining structural controllability conditions with the already known observability structural conditions, efficient control algorithms are expected to be proposed for the local control, leading to the possibility of efficiently controlling more $TCPN$s subclasses in both centralized and distributed approaches.

CRediT authorship contribution statement

J.L. García-Malacara: Conceptualization, Methodology, Formal analysis, Writing – original draft. **César Arzola:** Conceptualization, Methodology, Formal analysis, Writing – original draft. **Antonio Ramírez-Treviño:** Conceptualization, Methodology, Formal analysis, Writing – original draft. **C. Renato Vázquez:** Conceptualization, Methodology, Formal analysis, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix. Additional results

Proposition 13. *Let $\langle \mathcal{N}, \lambda\rangle$ be a $TCPN$ where $\mathcal{N}$ is a consistent and strongly connected Choice-Free net. If $\langle \mathcal{N}, \lambda\rangle$ is rank-controllable, then for every P-subnet $\mathcal{P}_y$, induced by the P-semiflow $\boldsymbol{y}$, $\exists t \in T_c$ such that $t \in \mathcal{P}_y$.*

Proof. Since $\mathcal{N}$ is strongly connected Choice-Free, then it is conservative. Thus every place belongs to the support of a minimal P-semiflow. Moreover, each minimal P-semiflow $\mathbf{y}$ induces a P-subnet, $\mathcal{P}_y$, that is a strongly connected $FA\ PN$.

Suppose that there exists a $\mathcal{P}_y$ with k places and no controllable transition. Let P' and T' be the places and transitions of $\mathcal{P}_y$. Let C_i be any configuration where the places in P' are constraining their respective output transitions. Rename the places and transitions in $\langle \mathcal{N}, \lambda\rangle$ in such a way that the places and transitions in $\mathcal{P}_y$ are the first k places and transitions, respectively. Then, the incidence matrix is rewritten as:

$$\boldsymbol{C} = \begin{bmatrix} \boldsymbol{C}_{11} & \boldsymbol{C}_{12} \\ \boldsymbol{C}_{21} & \boldsymbol{C}_{22} \end{bmatrix}$$

where $\boldsymbol{C}_{11} = \boldsymbol{C}(P', T')$, $\boldsymbol{C}_{12} = \boldsymbol{C}(P', T \setminus T')$, $\boldsymbol{C}_{21} = \boldsymbol{C}(P \setminus P', T')$, and $\boldsymbol{C}_{22} = \boldsymbol{C}(P \setminus P', T \setminus T')$. Notice that $\boldsymbol{C}_{12} = \mathbf{0}$, otherwise, some transitions in $T \setminus T'$ should be part of $\mathcal{P}_y$, which is a contradiction. Hence, matrix $\boldsymbol{\Pi}_i$ may be rewritten as:

$$\boldsymbol{\Pi}_i = \begin{bmatrix} \boldsymbol{\Pi}_{i11} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Pi}_{i22} \end{bmatrix}$$

where $\boldsymbol{\Pi}_{i11}$ is a $k\times k$ non-singular matrix relating places to transitions of $\mathcal{P}_y$, and $\boldsymbol{\Pi}_{i11}$ is related to the places constraining transitions not in $\mathcal{P}_y$.

By appropriately partitioning matrix $\boldsymbol{\Lambda}$, the dynamic matrix can be rewritten as:

$$\boldsymbol{C}\boldsymbol{\Lambda}\boldsymbol{\Pi}_i = \begin{bmatrix} \boldsymbol{C}_{11} & \mathbf{0} \\ \boldsymbol{C}_{21} & \boldsymbol{C}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Lambda}_{11} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Lambda}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Pi}_{i11} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Pi}_{i22} \end{bmatrix} = \begin{bmatrix} \boldsymbol{C}_{11}\boldsymbol{\Lambda}_{11}\boldsymbol{\Pi}_{i11} & \mathbf{0} \\ \boldsymbol{C}_{21}\boldsymbol{\Lambda}_{11}\boldsymbol{\Pi}_{i11} & \boldsymbol{C}_{22}\boldsymbol{\Lambda}_{22}\boldsymbol{\Pi}_{i22} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11} & 0 \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}$$

Notice that $\mathbf{A}_{11}$ is a sub-matrix of dimension $k \times k$, and has one left eigenvector v_1 associated to $\mathbf{y}$ whose eigenvalue is zero, and has other $k-1$ left eigenvectors $\mathbf{v}_i$, that can be real, complex and generalized eigenvectors, however, each generalized eigenvector has an original eigenvector, then there is at least another left eigenvector. Moreover $\mathbf{eg}_i = [\mathbf{v}_i \quad \mathbf{0}]$ are left eigenvectors of $\boldsymbol{C\Lambda\Pi}_i$.

Finally, since the input matrix $\boldsymbol{C}_c$ has the form $[\mathbf{0} \quad *]^T$ ($\mathcal{P}_y$ has no controllable transitions) and it is orthogonal to any eigenvector $\mathbf{eg}_i$, then $\mathbf{eg}_i \neq \mathbf{y}$ is in the kernel of $\mathscr{C}_i$ then the net is not rank-controllable in this configuration. In fact, the dynamic of the marking of places in the P-semiflow in this configuration is independent of the input. ■

References

[1] A. Pózna, A. Fodor, M. Gerzson, K. Hangos, Colored Petri net model of electrical networks for diagnostic purposes, IFAC-PapersOnLine 51 (2) (2018) 260–265.

[2] H.Z. Nabi, T. Aized, Performance evaluation of a carousel configured multiple products flexible manufacturing system using Petri net, Oper. Manage. Res. 13 (1) (2020) 109–129.

[3] C. Avram, J. Machado, A. Aştilean, Methods for modeling urban road traffic using timed automata, in: Design, Simulation, Manufacturing: The Innovation Exchange, Springer, 2020, pp. 97–107.

[4] H. Qi, Y. Yu, Q. Tang, X. Hu, Intersection traffic deadlock formation and its probability: A Petri net-based modeling approach, IET Intell. Transp. Syst. (2022).

[5] H. Kaid, A. Al-Ahmari, Z. Li, R. Davidrajuh, Single controller-based colored Petri nets for deadlock control in automated manufacturing systems, Processes 8 (1) (2019) 21.

[6] L. Li, F. Basile, Z. Li, An approach to improve permissiveness of supervisors for GMECs in time Petri net systems, IEEE Trans. Automat. Control 65 (1) (2019) 237–251.

[7] X. Cong, M. Fanti, A. Mangini, Z. Li, Critical observability of partially observed Petri nets, IFAC-PapersOnLine 53 (4) (2020) 350–355, 15th IFAC Workshop on Discrete Event Systems WODES 2020 — Rio de Janeiro, Brazil, 11-13 November 2020.

[8] H. Lan, Y. Tong, C. Seatzu, Analysis of strong and strong periodic detectability of bounded labeled Petri nets, Nonlinear Anal. Hybrid Syst. 42 (2021) 101087.

[9] R. David, H. Alla, Discrete, Continuous, and Hybrid Petri Nets, Springer, Berlin, Heidelberg, 2010.

[10] M. Silva, J. Júlvez, C. Mahulea, C.R. Vázquez, On fluidization of discrete event models: observation and control of continuous Petri nets, Discrete Event Dyn. Syst. 21 (4) (2011) 427–497.

[11] M. Dotoli, M.P. Fanti, A.M. Mangini, W. Ukovich, A continuous Petri net model for the management and design of emergency cardiology departments, IFAC Proc. Vol. 42 (17) (2009) 50–55.

[12] J. Júlvez, R. Boel, A continuous Petri net approach for model predictive control of traffic systems, IEEE Trans. Syst. Man Cybern. A 40 (4) (2010) 686–697.

[13] M. Beccuti, C. Fornari, G. Franceschinis, S. Halawani, O. Barukab, A. Ahmad, G. Balbo, From symmetric nets to differential equations exploiting model symmetries, Comput. J. 58 (1) (2013) 23–39.

[14] J. Xu, L. Xie, Control and Estimation of Piecewise Affine Systems, Elsevier, 2014.

[15] E. Aguayo-Lara, A. Ramírez-Treviño, J. Ruiz-León, Sensor placement for distinguishability and single structure observer design in continuous timed Petri nets, Nonlinear Anal. Hybrid Syst. 12 (2014) 20–32.

[16] E. Aguayo-Lara, A. Ramírez-Treviño, J. Ruiz-León, Invariant subspaces and sensor placement for observability in continuous timed Petri nets, in: 2011 IEEE International Conference on Automation Science and Engineering, 2011, pp. 607–612.

[17] C. Arzola, C.R. Vázquez, M. Silva, A. Ramírez-Treviño, Structural characterization of controllability in timed continuous Petri nets using invariant subspaces, IFAC-PapersOnLine 53 (2) (2020) 2087–2094.
[18] J. Júlvez, E. Jiménez, L. Recalde, M. Silva, On observability and design of observers in timed continuous Petri net systems, IEEE Trans. Autom. Sci. Eng. 5 (3) (2008) 532–537.
[19] C. Mahulea, L. Recalde, M. Silva, Observability of continuous Petri nets with infinite server semantics, Nonlinear Anal. Hybrid Syst. 4 (2) (2010) 219–232.
[20] C.R. Vázquez, A. Ramírez-Treviño, Structural and generic conditions for controllability of timed continuous Petri nets, in: Proc. of the 11th International Workshop on Discrete Event Systems, 2012, pp. 41–46.
[21] C.R. Vázquez, A. Ramírez-Treviño, M. Silva, Controllability of timed continuous Petri nets with uncontrollable transitions, Internat. J. Control 87 (3) (2014) 537–552.
[22] C.R. Vázquez, J.L. García-Malacara, A. Ramírez-Treviño, Duality of controllability-observability in continuous Petri nets, in: Proceedings of the 19th IEEE International Conference on Emerging Technologies and Factory Automation, Barcelona, Spain, 2014.
[23] M.H.T. Hack, Analysis of Production Schemata by Petri Nets, Technical Report, Massachusetts Inst. of Tech. Cambridge Project MAC, 1972.
[24] E. Teruel, J. Colom, M. Silva, Choice-free Petri nets: a model for deterministic concurrent systems with bulk services and arrivals, IEEE Trans. Syst. Man Cybern. A 27 (1) (1997) 73–83.
[25] C.-T. Chen, Linear Systems Theory and Design, Saubders College Publishing, Philadelphia, PA, USA, 1984.
[26] J. Júlvez, L. Recalde, M. Silva, Steady-state performance evaluation of continuous mono-T-semiflow Petri nets, Automatica 41 (4) (2005) 605–616.
[27] C.R. Vázquez, M. Silva, Timing-dependent boundedness and liveness in continuous Petri nets, IFAC Proc. Vol. 43 (12) (2010) 7–14.